Chapter 3: Complications and resolutions

INTRODUCTION

In Chapter 2, we valued some simplified cash flow patterns commonly seen in advanced financial engineering practice, but we made several simplifying assumptions.

- We assumed that the discount rate was constant, interest was compounded once per period, the first cash flow occurred in one period, and that the remaining cash flows were known and riskless.

We need to adjust for all types of payment frequencies and discounting or yield conventions.

- Cash flow valuations in practice are affected by inflation and also by taxes; hence we must be able to adjust our valuations to properly reflect these adjustments.
- Events may cause our cash flows to suddenly change or even cease, for example, a default by the payer, the death of the receiver, or a contract provision triggering a windfall or an early termination.
- We would like to be able to value cash flows linked to fluctuating interest rates or inflation rates.
- We should know how to perform valuations in alternative currencies.

These are the subjects of Chapter 3. The major element missing from the list of concerns above is the proper treatment of risk. This will be forthcoming in Part II. For now, we will make some adjustments for expected losses due to risk, but address the pricing of risk in later chapters. In some special cases, however, we are fortunate enough to know a *risk-neutral* equivalent price, which has a risk adjustment embedded, so that the risk-neutral price can be discounted at the riskless rate to find its present value. We can use risk-neutral pricing to determine the value of the individual payments in floating rate bonds, and the value of foreign-currency denominated cash flows. Risk-neutral pricing will become increasingly important in later chapters.

To help you understand all the rate adjustments in this chapter, we offer you the following lexicon as a reference:

RATE	DEFINITION
Average tax rate	Total taxes divided by total income, expressed as a percentage
Effective annual rate	The rate of interest earned or paid for a year considering the effect of periodic compounding and earning interest on interest
Forward rates	The year-by-year implied future returns embedded in a bond price.
Foreign exchange forward rate	The pre-agreed exchange rate for a contracted currency exchange in the future
Inflation rate	The percentage increase in the value of a fixed basket of goods over a period of time, typically one year. Inflation also reflects the decline in purchasing power per year of a fixed amount of money.
Marginal tax rate	The amount of tax paid on an incremental dollar of income, expressed as a percentage
Nominal rate	The discount rate used to value actual dollars paid (ignoring inflation)
Periodic rate	The simple annual rate divided by the number of periods a year
Real rate	The discount rate used to value inflation-adjusted cash flows, or the rate of growth in spending power of savings

Simple annual rate	The discount rate used to discount annual cash flows, i.e. as if interest were compounded once a year
Spot rate	The annualized rate of return on a bond bought today and held to maturity.

3.1 CHANGING THE FREQUENCY OF PAYMENTS

I early inquired the rate of interest on invested money, and worried my child's brain into an understanding of the virtues and excellencies of that remarkable invention of man, compound interest.

- Jack London, 1906

In Chapter 2, we assumed all payments were annual, and were discounted at the simple annual interest rate. Suppose we have payments every month, like consumer loans? Or every quarter (3 months), like some corporate bonds? Or every half-year, like some corporate bonds and all U.S. government bonds? How are valuations affected? In practice, as a matter of convention, we usually simply divide the quoted simple annual rate by the number of periods per year. Therefore, if "r" is the quoted simple annual rate, a monthly time step would imply a simple monthly interest rate of r/12. A 6% interest rate on a loan with monthly payments therefore has a monthly interest rate of 0.50% or 50 basis points per month (bppm).

At first, this concept may puzzle some students. If we invest $1,000,000 now and reinvest the 0.5% per month for 12 months at the same rate, the future value in one year would be $\$1{,}000{,}000 \times (1.005)^{12} = \$1{,}061{,}678$, which *effectively* amounts to 6.17% over the year, well over 6% per year thanks to the effect of compound interest. That is because interest is being paid on the interest. This seems like the more honest calculation than simple interest, since it considers the return using actual compounding. Multiplying and dividing an interest rate by a constant seems intuitively incorrect, even though it is simple, convenient and a widely used convention. It is also not necessarily the case in practice that interest can be reinvested at the same return that the principal earns.[6]

It is important to realize that in almost every capital market, there are different legacy quoting conventions for reporting prices and rates of return, such as the simple annual rate, but usually some adaptation needs to be made by investors in order to compare effective rates of return between investments. To help investors or borrowers distinguish between the quoted rate and the actual earned return, we often call the compounded return the *effective annual yield* or *effective annual rate*.

Quick quiz 1: If the simple annual rate on a loan is 8%, and interest is compounded quarterly, what is the effective annual rate? (Ans: $8.24\% = (1+0.08/4)^4 - 1$)

Quick quiz 2: If the effective annual rate is 12.55%, and the payment frequency is quarterly, what was the simple annual rate? (Ans: 12%)

Quick quiz 3: What assumptions are made in calculating the effective annual rate? (Ans: The instrument is held to maturity, and interest payments are reinvested at the same yield as the principal.)

There is another way to remember this easily. For example, when you consider the yield to maturity of a bond, also known as the *bond equivalent yield (BEY)*, the reported yield is not the same as the effective yield. However, the semi-annual yield *(BEY/2)* equals the effective yield on the bond per six-month period on the

[6] This risk is also known as *reinvestment risk*.

assumption that the bond is held to maturity and the coupons are reinvested at the same rate as the principal. It is only when we annualize the returns that we have to compound returns on assets with different payment frequencies.

To summarize in an equation, if r is the simple annual interest rate, EAY is the effective annual yield, and k is the number of compounding periods per year:

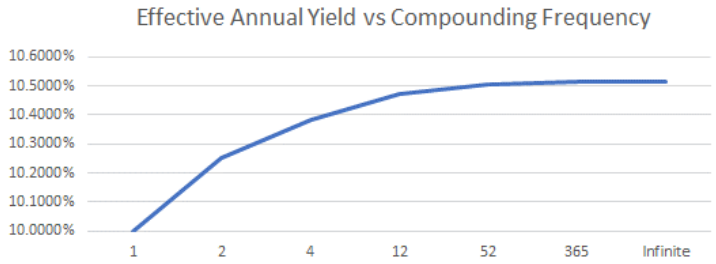
$$1 + EAY = (1 + r/k)^k$$

The effective annual rate is rarely computed in practice when simple rates can be compared among similar instruments. However, it is a necessary tool in careful return analysis, particularly when comparing the returns of assets with different payment frequencies.

Continuous compounding

As we saw in the last section, increasing the compounding frequency to 2, 4 or 12 times a year increases the effective annual return. What would happen if we increased to 52, 365, or even an infinite number of times per year?

Frequency	EAY
1	10.0000%
2	10.2500%
4	10.3813%
12	10.4713%
52	10.5065%
365	10.5156%
Infinite	10.5171%



This exhibit shows the effective annual yield for a 10% simple annual rate at different compounding frequencies on a nonlinear scale. The maximum rate that can ever be attained is 10.5171%. As the number of compounding periods tends to infinity, the discrete model is said to become *continuous*. We will refer to the 10% figure as the continuously discounted rate, and continue to refer to 10.5171% as the effective annual yield in this case.

In general, we may continue to interpret r as the simple annual interest rate, but $e^r - 1$ is the effective annual rate when the number of compounding periods is infinite. This can be derived by taking the limiting value of a $1 investment in one year, and compounding n times a year. As n goes to infinity, the future value of $1 is given by[7]:

$$\lim_{k\to\infty}\left(1+\frac{r}{k}\right)^k = e^r$$

This means that the present value of any payment C_t at time t, discounted continuously at the simple annual rate of r is given by

[7] This can be verified by applying L'Hôpital's rule in computing the limit of the natural log.

$$PV = C_t e^{-rt}$$

In many applications, especially when we study fixed income and derivatives, you may see all discount rates converted to their continuously compounded equivalents. Like the EAY, this has the benefit of eliminating differences in yield calculations caused by cash flows being discounted at different periodic rates.

Fortunately, there is an easy conversion between effective annual yield and the continuously compounded simple interest rate:

$$r = ln(1 + EAY)$$

Exercise

If an investment offers 3% simple return per half-year, what is the equivalent continuously compounded discount rate?

Solution

(Ans: The value of $1 + EAY$ is 1.03^2. The continuous yield is $5.91\% = \ln(1.03^2)$)

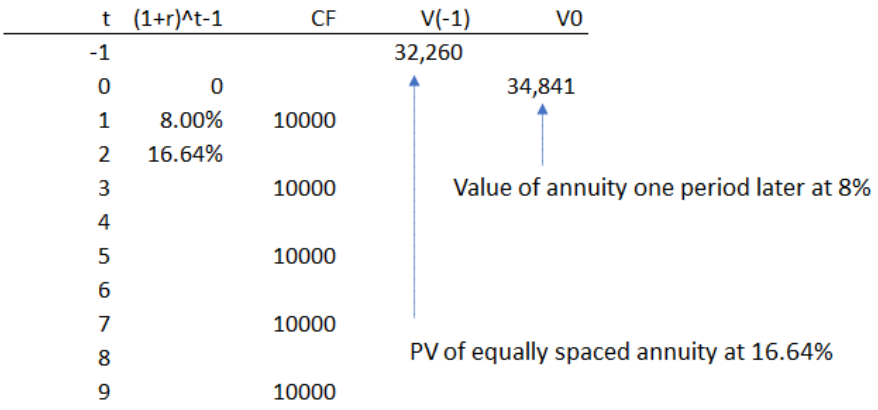
In practice, continuously compounded rates are rare in corporate finance, but are used extensively in valuations involving fixed income and exclusively when valuing derivatives. For this reason, the financial engineering student must be very comfortable with continuous discounting formulae and procedures. Also, for analytical purposes, it is often most efficient to use continuous discounting for discrete cash flows.

Payment frequency and discounting frequency

Payment frequency refers to the number of payments per period. *Discounting frequency* refers to the number of periods used per period to discount cash flows. When you encounter problems with different payment frequency and discounting frequency, it is best to begin by computing the discount rate corresponding to the payment frequency. For example, the interest rate on consumer loans is quoted on an annual basis, but payments are made on a monthly basis. In this case, we divide the simple annual rate by 12 in order to get the monthly rate, understanding that this is the standard rate convention --- ignoring the temptation to "de-compound" the return.

Now suppose we have a biennial payment (once every two years) of $10,000 starting in one year, repeating every second year for 5 payments. If the simple annual discount rate is 8%, what is the present value? We'd like to use the annuity formula to find the present value, but there are two problems. First, the quoted interest rate is annual, while payments occur every two years. Second, the first payment is in one year, which is in one-half of a two-year time period. To solve this problem, we first compute the effective two-year discount rate, which is $1.08^2 - 1$ or 16.64%. Then we find the PV of the annuity, or $32,260. But this is the present value *a year ago,* one period (one half of a two year period) before the first payment in one year. Hence we must multiply by 1.08 to find the present value today, or $34,841.

The Excel presentation summarizes this calculation:



This section discussed how to compute discount rates for different compounding periods and starting dates. Because consumer loans make monthly payments, corporate loans make quarterly or semiannual payments and government loans make semiannual payments, financial analysts frequently need to be able to convert yields from one compounding frequency to another. Also, for investment purposes, analysts need to convert simple rates of return into effective rates in order to compare different investments. As noted, this means that we are assuming the payments are reinvested at the same rate as the principal, and that the bonds are held to maturity.

In sum, the usual convention is to take a simple annual rate and divide by the number of compounding periods per year, ignoring the effects of compound interest. Then, an effective annual yield may be determined considering the effect of compounding. Finally, continuously compounded rates are sometimes used as a uniform discounting standard that allows analysts to compare yields at different compounding frequencies on an equivalent basis, even when payments are not continuous.

3.2 INFLATION

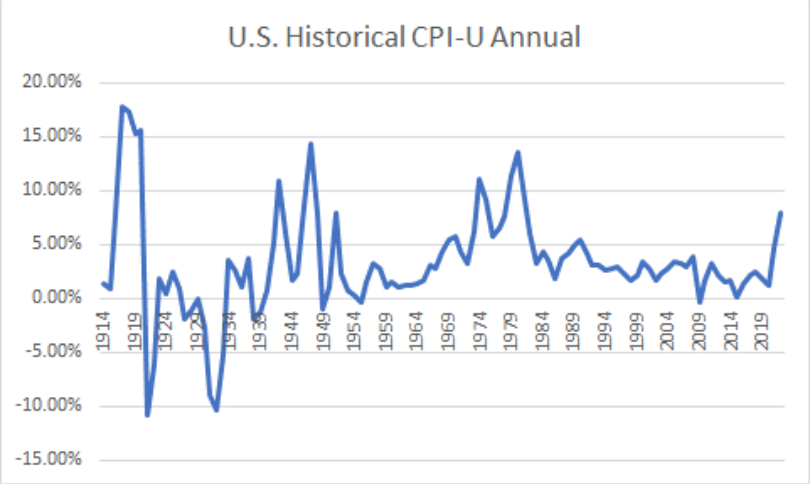
"I was reading in the paper today that Congress wants to replace the dollar bill with a coin. They've already done it. It's called a nickel."
— Jay Leno

"*The arithmetic makes it plain that inflation is a far more devastating tax than anything that has been enacted by our legislature. The inflation tax has a fantastic ability to simply consume capital. It makes no difference to a widow with her saving in a 5 percent passbook account whether she pays 100 percent income tax on her interest income during a period of zero inflation, or pays no income taxes during years of 5 percent inflation. Either way, she is 'taxed' in a manner that leaves her no real income whatsoever. Any money she spends comes right out of capital. She would find outrageous a 120 percent income tax, but doesn't seem to notice that 5 percent inflation is the economic equivalent."*
— Warren Buffett

Until now, we have ignored inflation. In reality, the purchasing power of a currency generally declines over time, so that the value of $1 today does not buy as many goods in 10 years. The rate of decline in purchasing power is also the rate of inflation of the currency. For example, if a can of beans cost $1 ten years ago, and it costs $1.50 today to buy the same can of beans, then the inflation rate, as measured by beans would be $(1.50/1.00)^{1/10} - 1 = 4.14\%$ per year. It is also possible that a currency may *deflate*, such as in the U.S. Great

Depression of 1929-33, indicating a negative rate of inflation. In this unusual situation, the spending power of a currency increases over time.

In the 2010's in the U.S., the rate of inflation averaged just under 2% per year. Inflation is commonly measured by the CPI-U (Consumer Price Index for All Urban Consumers), which is computed by the U.S. Bureau of Labor Statistics. The CPI-U is computed as a price index over time, although inflation is normally reported as the year-on-year change in the CPI-U. In 2022 for example, the CPI-U inflation rate was 8%. The chart below shows a historical perspective. In the 1970s and early 1980s, the U.S. experienced its highest modern inflation rates, both high in level and in volatility. Since then, rates fell and become more stable, at least until COVID arrived in 2020.



Source: Minneapolis Federal Reserve Bank

The CPI-U includes consumer costs in the following categories: Food & beverages, housing, apparel, transportation, medical care, recreation, education, communication, and other goods & services. The CPI-U includes taxes such as sales taxes that are associated with consumer expenses, but excludes other taxes. It excludes all investment related items.

Of course, the CPI-U will not in general be able to match anyone's actual increase in expenses, since some will necessarily spend relatively more on medical care vs education and vice versa. Nevertheless, it is considered close enough for most consumers. In fact, many consumer payments, such as cost-of-living adjustments to wages (so-called *COLA* or inflation-protection clauses), and pension benefits are linked to the CPI. If your wages are *indexed* to the CPI, this means your actual cash income will rise in proportion with the CPI. If you make $50,000 in 2017, your base wage does not increase, you have a COLA clause in your wage contract, and the CPI inflation rate ends up being 2%, you will nominally earn $51,000 in 2018, but your spending power will be $50,000 in 2017 dollars.

As an index choice, the CPI does not work in all situations. There are custom regional indices available for example, which you might use to negotiate a COLA contract specific to Syracuse or Seattle. There are also indexes related to producer prices (PPI), since manufacturing companies purchase baskets of industrial commodities that differ significantly from consumer baskets. Finally, some analysts prefer to use the GDP

(gross domestic product) deflator, which necessarily considers all goods and services produced in an economy as the "basket", and measures the rate of price increase of that basket. The GDP deflator better reflects the ability of USD to purchase a pro-rated basket of total U.S. production of goods and services, as compared to the previous year.

Can inflation be a good thing?

We've been considering the effect of inflation on wage-earners, who would be hurt by inflation if they received fixed wage payments while inflation was present. However, if the same wage-earner were making fixed payments on a mortgage while his salary was indexed to inflation, he would be pleased to see his purchasing power increasing on a relative basis rather than decreasing.

Nominal and real cash flows and rates

Whatever inflation index you use going forward, we will need to differentiate between *nominal* cash flows and *real* cash flows. Real cash flows have been adjusted for inflation so that the purchasing power of a real dollar in one year is the same as another year. For example, if your nominal income went from $50,000 in 2022 to $51,000 with 2% inflation, your real income in terms of constant spending power stayed the same at $50,000 in 2022 dollars.

The valuations we have done so far have all been based on nominal cash flows, indeed in practice, almost all valuations are like this, since cash payments are normally expressed without any inflation adjustment. Income and expenses are estimated in nominal dollars, and are discounted at nominal interest rates. Whenever we use "r" in this text, it represents the nominal rate of interest.

Real cash flows are those that have been adjusted for spending power. With inflation present, the spending value of a nominal dollar will depend on the inflation rate. Real dollars need to be discounted at a different rate of interest. We will use the Greek letter rho "ρ" to denote the real rate of interest. Also, "i" is the inflation rate.

The famous **Fisher Equation** links one period inflation to one period spending power. Interest raise the dollar's spending power in the future (multiplying by (1 + r)), while inflation reduces it (dividing by (1 + i)). The resulting ratio expresses your real increase in purchasing power (1 + ρ).

$$\textbf{\$1 for 1 yr} \rightarrow \textbf{(1+r)/(1+i)} \equiv \textbf{(1+}\rho\textbf{)}$$

$$or\ \rho = (r - i)/(1+i)$$

With continuously discounted inflation and interest rates, we have exactly.

$$e^r = e^{\rho}e^i \quad or \quad r = \rho + i$$

This is easily computed by taking natural logs of both sides. For example, if the interest rate is 5%, but the inflation rate is 2%, a risk-free investment increases in spending power by about 3% per year. One nominal dollar now has the same real value in today's dollars, unless we define it otherwise using a historical benchmark, such as "1960 dollars". After a year, a nominal dollar is worth $1.05, but its spending power is about $1.03 in today's dollars.

To value real annuities, therefore, we must adjust the growth rate to reflect a currency's decline in spending power. However, we have already solved this problem analytically in Chapter 2. Let the cash flow growth rate $g = -i$ to convert nominal cash flows to real cash flows. Then apply a real discount rate to the real cash flows.

As an example, let's consider a COLA wage annuity that pays an inflation-linked cash flow, with C_0 being the base income, $C_1 = C_0(1+i)$ being the first nominal payment in one year, $C_2 = C_0(1+i)^2$, and so on.

We can easily value this as a growing annuity with g = i

$$V_0 = \frac{C_0(1+i)}{(r-i)}\left[1 - \frac{(1+i)^n}{(1+r)^n}\right]$$

But if we let $r^* = (r - i)/(1+i)$, we have the regular annuity formula

$$V_0 = \frac{C_0}{r^*}\left[1 - \frac{1}{(1+r^*)^n}\right]$$

Which is exactly the simplification we learned in Chapter 2 for valuing exponentially growing annuities. Of course r* is actually ρ in this case. If ρ=0, we can simply add up the real cash flows to determine the value

$$V_0 = nC_0 \qquad if\ r^* = 0$$

This relationship is particularly important to understand when inflation rates are unknown. In this case, it would be very difficult to forecast nominal dollars and know what rate to use to discount them. We could however assume that the real payment was the same in every year, and that the cash flows were a constant annuity, to be discounted at the rate of ρ.

Example

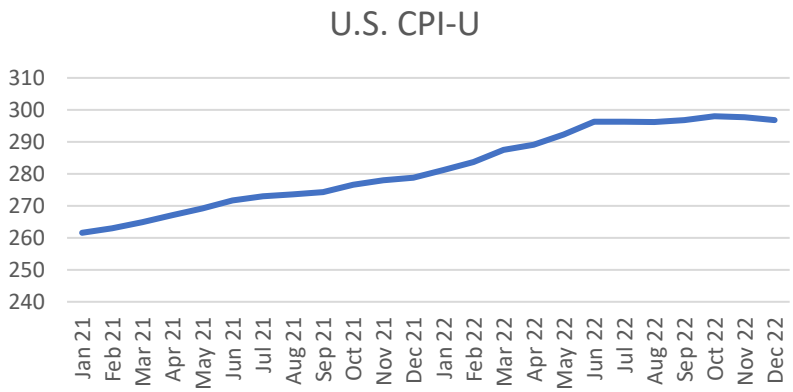
Paul earned $100,000 in the past year. He just signed a 3-year employment contract with a COLA clause. His base salary will be adjusted upward in the future according to the CPI, currently 2% per year. If the nominal rate of interest is 4.5%, what is the present value of the contract revenues?

Solution

$286K; find the present value of three years of constant base pay at the real rate of interest.

Recent CPI-U values (through 2022):

The graph below shows the historical values of CPI-U for 2021 and 2022. This was a time of high inflation concerns, considering the COVID expenses, other high government expenditures and the Ukrainian War. The top graph shows the price of the CPI-U basket leveling off, suggesting that inflation is approaching zero. However, since inflation is usually computed as a percentage change from the previous year, the accompanying table below shows that the reported CPI inflation rates appear to be falling.



Mo/Yr	2022	2021	%Δ
Jan	281.148	261.582	7.48%
Feb	283.716	263.014	7.87%
Mar	287.504	264.877	8.54%
Apr	289.109	267.054	8.26%
May	292.296	269.195	8.58%
Jun	296.311	271.696	9.06%
Jul	296.276	273.003	8.52%
Aug	296.171	273.567	8.26%
Sep	296.808	274.31	8.20%
Oct	298.012	276.589	7.75%
Nov	297.711	277.948	7.11%
Dec	296.797	278.802	6.45%

www.rateinflation.com

Inflation rate calculation

Dec 2022 CPI-U: 296.797
Dec 2021 CPI-U: 278.802
Inflation rate: 296.797/278.802 – 1 = 6.45%

If you are fortunate enough to have a cost-of-living-adjustment (COLA) clause in your wage contract, this means that your future wage will be determined by the formula:

Future wage = Base wage × Future CPI index level / Starting CPI level

Of course, actual inflation is unknown ahead of time, but COLA clauses protect people from unforeseen increases in the inflation rate as defined by the CPI.

In sum, the most important things to know about inflation are:

- ✓ The CPI-U is the most popular inflation index used in the U.S., but it is not suited to all purposes. It would be impossible to create a single index to measure everyone's change in purchasing power. Regional CPI indices (for employers), the PPI (for manufacturers) and the GDP deflator (for investors) are the most popular alternatives.

- ✓ *Nominal* cash flows are actual dollar cash flows, while *real* cash flows are adjusted for inflation. Real cash flows are expressed in a previous year's dollar value, e.g. 1960 dollars.
- ✓ Discount nominal or real cash flows at their respective interest rates r and ρ, never mix them.
- ✓ The growing annuity tools can be used to value real and nominal cash flows.
- ✓ $\rho = (r - i)/(1+i)$, nearly equal to $r - i$ for small i. This is exact with continuous discounting: $\rho = (r - i)$
- ✓ In today's dollars, the value of nominal cash flows discounted at nominal rates equals the value of real cash flows discounted at real rates.

The last point may be the most important one. It may seem difficult to value a stream of cash flows linked to an unpredictable future CPI index level. However, if the cash flows are inflation-adjusted, we can discount the future payments expressed in today's dollars at the real rate of interest. As we will see in Chapter 4, the real rate of interest may be estimated using the yield to maturity of inflation protected bonds (TIPS).

3.3 TAXES

One year while I was at his Princeton home preparing his return, Mrs. Einstein asked me to stay for lunch. During the course of the meal, the professor turned to me and with his inimitable chuckle said: "The hardest thing in the world to understand is income taxes." I replied: "There is one thing more difficult, and that is your theory of relativity." "Oh, no," he replied, "that is easy." To which Mrs. Einstein commented, "Yes, for you."

LEO MATTERSDORF New York City

In real life, taxes are complicated. Nevertheless, taxes can play a pivotal role in valuations and in investment selections. Governments use tax rules to encourage people to invest in government-favored assets, increasing the value of some assets relative to others. Individuals seek alternative investments to reclassify or offset income in order to reduce their taxes. A precise understanding of the taxes in any activity will be important for you to determine accurate valuations and appropriate investment strategies. However, your private valuations will depend on your tax situation. It will be impossible to cover all the complications of taxes here. Simulation methods covered in Chapters 5 and 6 can deal fairly easily with complex tax schemes, even when it might be difficult to develop analytic pricing formulas.

Let's begin with some terminology. *Nominal tax rates* refer to the stated tax rate for a given situation. *Effective tax rates* refer to the rate actually paid. Taxes may be levied on capital gains and cash flows, usually both, albeit at different marginal tax rates. The percentage tax paid on an incremental dollar of income is called the *marginal tax rate*. Total taxes divided by the investment amount is called the *average tax rate*. Taxes may be assessed on capital gains, but often, tax is only due in the tax period the capital gain is *realized*, which occurs by definition when the asset is sold or as required by the applicable tax code. For example, in mark-to-market accounting, taxes are computed based on the change in market value of an asset regardless of whether it was sold or not.

While it is impossible to list all the complications surrounding real life taxes, here is a list of important things to know:

Realistic assumptions:

- Capital gains are generally taxed when they are realized, on the sale of an asset
    - Capital losses are deducted from taxes, but must also be realized before they are taxed

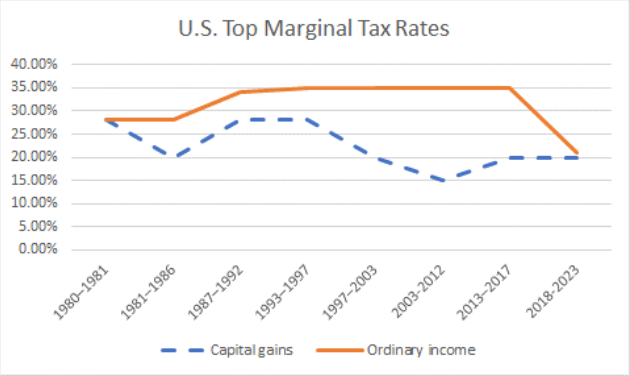
    - There are generally limits to allowable deductions
    - Losses may be allowed or required to be *carried forward*, i.e. recognized in future periods
- In some cases, accrued but unrealized capital gains may be taxable (e.g. mark-to-market accounting)
- Capital gains tax rates differ from ordinary income (cash) tax rates
- Tax rates are progressive, meaning that people with higher incomes have higher marginal tax rates
- Tax rates differ for different individuals and corporation types
- Tax rates change over time

For the purposes of this chapter, we will make the simplifying assumptions as follows:

- Capital gains and losses are taxed at the time they occur (i.e. they are all treated as realized)
- Tax rates are constant, and do not change based on income or ownership
- Loss deductions are unlimited

For the moment, we will allow capital gains tax rate (τ_{CG}) to differ from the ordinary income tax rate (τ_{OI}). A rule of thumb in finance states that you may discount pretax cash flows at pretax interest rates, or after tax (AT) cash flows at after tax interest rates. These calculations should yield the same present value, and in many cases, this equivalence defines the *effective* after-tax discount rate. When capital gains (CG) and ordinary income (OI) are taxed at the same rate τ, the after-tax of return $r_{AT} = r(1 - \tau)$, if r is the pretax rate of return.

Historically the two tax rates have generally differed in the U.S. since 1922, with the top income tax rate exceeding the capital gains rate. However, they are equal in 2018-23. Here are the two rates from 1980-2023:



Source: Wolters Kluwer CCH

Example

A par bond is one whose value is the same as its principal, often the case on the issuance date. Consider a par bond with annual coupons, which costs $1000 and pays annual coupons at a rate of 5% ($50 per year) until maturity. At maturity, the $1000 principal is repaid along with the final coupon. If the tax rate $\tau = 40\%$, then the bondholder only keeps $30 from the coupon payment, which is $(1 - \tau)$ or 60% of the $50, since $20 is paid in taxes. There is no capital gain to tax, since the bond is redeemed for $1000 at maturity.

The pretax present value is $1000 if the interest rate is also 5%. The after-tax cash flows resemble a pretax bond paying $30 per year and $1000 at maturity, so the realized rate of return, $r_{AT} = 3\%$. Indeed, in this case the equation holds --- $3\% = (1 - 0.40) \times 5\%$.

The effect on the GVE of taxes is simple in theory. If τ is the tax rate, and capital gains and cash flows are taxed at the same rate, and our assumptions from above hold, then after-tax required and expected returns are $(1 - \tau)$ times the pretax returns in dollars:

Ignoring risk, when the capital gains rate is the same as the ordinary income rate, the GVE states,

$$\text{Pretax required return} \times (1 - \tau) = \text{Pretax capital gain} \times (1 - \tau) + \text{Pretax cash flows} \times (1 - \tau)$$

Since all the components are multiplied by $(1 - \tau)$, we can see that the valuation equation is not affected by the taxes under our assumptions. If there is only one tax rate, then $(1 - \tau)$ cancels in the GVE. One may either express the GVE in pretax or post-tax terms, and arrive at the same valuation.

Different tax rates can also be incorporated into the GVE. Again ignoring risk,

$$\text{Pretax required return} \times (1 - \tau_e) = \text{Pretax capital gain} \times (1 - \tau_{CG}) + \text{Pretax cash flow} \times (1 - \tau_{OI})$$

This equation assumes that all capital gains are taxed at the same constant rate τ_{CG}, all cash income (or losses) are taxed at a potentially different constant rate τ_{OI}. The after-tax required return is determined from the pretax return and the effective tax rate is τ_e. Implicitly, this representation of the valuation assumes any losses are fully deductible from taxes paid --- negative taxes would reduce an investor's tax burden or even lead to a refund. For example, if one had to defer capital losses, then "$(1 - \tau_{CG})$ Capital gains" could be replaced with "$(1 - \tau_{CG})$ Max(Capital gains, 0)".

Here is a numerical example for a 10-year annuity of \$50 per year, and r=5%, and both tax rates equal to 40%:

	ACTUALS			TAX			
Time	CF	V	CG	CF	CG	Total	AT
0		386.09					
1	50.00	355.39	(30.70)	20.00	(12.28)	7.72	42.28
2	50.00	323.16	(32.23)	20.00	(12.89)	7.11	42.89
3	50.00	289.32	(33.84)	20.00	(13.54)	6.46	43.54
4	50.00	253.78	(35.53)	20.00	(14.21)	5.79	44.21
5	50.00	216.47	(37.31)	20.00	(14.92)	5.08	44.92
6	50.00	177.30	(39.18)	20.00	(15.67)	4.33	45.67
7	50.00	136.16	(41.14)	20.00	(16.45)	3.55	46.45
8	50.00	92.97	(43.19)	20.00	(17.28)	2.72	47.28
9	50.00	47.62	(45.35)	20.00	(18.14)	1.86	48.14
10	50.00	0.00	(47.62)	20.00	(19.05)	0.95	49.05

In each time period, the annuity value is determined as the PV of the remaining payments. The capital gain (CG) is the change in value, which is negative for its life. The tax is 40% of the payment minus a credit for the capital loss. The after-tax cash flow (AT) is the pretax cash flow (payment) minus taxes. In this case, as expected, the present value of the after-tax cash flows at r_{AT} = 3% is \$386.09, exactly as in the pretax case.

What if $\tau_{CG} \neq \tau_{OI}$? To value an asset in this situation, the problem is that we don't know the value of r_{AT} from the outset. To solve the after-tax problem, we therefore take the following steps:

- ✓ Compute the value of the pretax instrument V_0 at pretax discount rate r and set this equal to the after-tax value
- ✓ Find the discount rate r_{AT} that makes the present value of the after tax cash flows equal to this value.
- ✓ The effective tax rate τ_e satisfies $r_{AT} = (1 - \tau_e)r$.

Example

A risk-free security sale is expected to generate $20 in dividends, $30 in capital gains, and a $1000 return of principal in one year. The security's value today is $1000, implying a 5% risk-free discount rate. The tax rate for ordinary income is 40%, and the capital gains rate is 15%. What is the effective tax rate and the after-tax rate of return on the security?

Solution

The pretax valuation is $1000, as is the post-tax value. The after-tax cash flows are $37.50 (= 0.6(20) + 0.85(30)) in one year. This suggests $r_{AT} = 3.75\%$ so $(1-\tau_e) = \frac{3.75}{5} = 0.75$, and the effective overall tax rate is 25%.

Finally, we may consider valuation with taxes to be something like a private valuation, in that investors with different tax situations will value the same set of pretax cash flows or investment returns differently. This may make certain investments, such as nontaxable municipal bonds, attractive for some investors and unattractive for others. In particular, nontaxable bonds will be preferred by investors in the highest tax brackets compared to investors in the lowest tax brackets.

In sum, it is difficult to make generalizations about valuations with taxes unless the capital gains and ordinary income tax rates are the same. In practice, it may be best to simulate after-tax outcomes of different investment programs to rank possible investments according to the tax circumstances for a particular investor. This implies that most tax-based valuations can be considered private valuations rather than public valuations of securities or contracts.

3.4 EVENT RISKS

Cash flows or capital gains may be subject to shocks at random times. If that is the case, we need to know how to value them. In this section, we demonstrate the application of the GVE to event risks when the probability of the occurrence of the event is constant in each period.

A life insurance policy

Consider the example of a single premium life insurance policy:

- V_0 = Value of the insurance policy now.
  The policy is purchased for V_0, and there are no recurring payments.
- C = Payoff at the time of death. The policy has zero residual value after the payment is made
- π = Probability of death each year, assumed constant.

For now, let's assume there is no risk premium. Then according to the GVE we have

$$\text{Required return} = rV_0$$

$$\text{Expected return} = \text{expected capital gain} + \text{expected cash flow} = -\pi V_0 + \pi C$$

The capital gain is $-\pi V_0$ because in any given year the policy is active, the possibility of terminating the policy is π and the capital loss is V_0. Similarly, the probability in any year of getting the cash flow is π so the expected cash flow is πC.

Setting these returns equal and solving for V_0

$$V_0 = \frac{\pi C}{r + \pi}$$

For example, suppose an insurance policy pays C = $100,000 at the time of death, and for a 30-year old, the conditional likelihood of death is 1% per year, at least for a while. The risk-free rate is assumed constant at 5%. Then the value of the insurance policy is $16,667 regardless of when it is bought, provided the probability of death remains constant. The resulting formula may be interpreted as a kind of perpetuity: every year, the expected payment is πC. We cannot discount at r since the policy will not last forever, but increasing the discount rate to $r+\pi$ automatically adjusts for the expected time to collect on the policy.

Now we may turn our attention to the risk premium. Under the assumptions made above, the insurance company would not make a return on the policy and could not recover any of the underwriting or management costs. If we assume the insurance company must earn a fraction k of the policy's value every year (let's say 0.5%), then the buyer must pay that total risk premium as follows:

$$\text{Required return} = rV_0 - kV_0 = (r - k)V_0$$

We could also add kV_0 to the right-hand side of the GVE, reflecting the fact that the insurance company earns part of the capital gain and cash flow of the investment. Whether we adjust the left or the right side of the GVE, to solve for V_0, we adjust the discount rate downward to 4.5%, and the cost of the policy rises to $100{,}000(0.01)/(0.045+0.01) = \$18{,}182$.

The increase in cost should not surprise anyone. In practice, sales, origination, administrative and risk and regulatory charges can be a significant part of the overall cost of any insurance policy.

Auto insurance

Consider the example of a single premium auto insurance policy:

- V_0 = Value of the insurance policy now.
    - The policy is purchased for V_0, and there are no recurring payments.
- C = Payoff in case of an accident. The policy remains in force after the accident and covers the policyholder for future accidents, until death.
- δ = Probability of death in any year, assumed constant.
- π = Probability of accident each year, assumed constant.
- k = Premium paid to the insurance company, as a percentage of the policy value

We assume for simplicity that the two events, death and accidents, do not interact with each other. We assume the occurrence of death or a single accident causes a payment at the end of the period. Then according to the GVE we have

$$\text{Required return} = rV_0 - kV_0$$

$$\text{Expected return} = \text{expected capital gain} + \text{expected cash flow} = -\delta V_0 + \pi C$$

The capital gain is $-\delta V_0$ because in any given year the policy is active, the possibility of terminating the policy is π and the capital loss is V_0. Similarly, the probability in any year of getting the cash flow is π so the expected cash flow is πC. The single premium value is therefore

$$V_0 = \frac{\pi C}{(r-k)+\delta}$$

If the automobile policy that will pay C=$15,000 for each year you are in an accident, the probability of an accident every year is π=10%, the risk-free rate is 5%, the death probability is 1% per year and the risk premium s 0.5%, then the value of the policy is $27,273.

Notice in this example, we can separate the probability of the cash event (in the numerator) from the probability of the termination event (in the denominator). The adjustment to the discount rate appears in the denominator due to the assumption of proportional fees & costs.

3.5 VARIABLE INTEREST RATES

In previous chapters, we assumed discount rates were constant over time. Of course, it is obvious from business news reports that interest rates are changing all the time. Even at a point in time, longer term discount rates will differ from shorter-term rates closer to the present. The graph of yield as a function of time or term is known as the *term structure* of interest rates. How can we determine the present value of cash flows defined by variable discount rates?

For expositional convenience, let's work with one-year intervals and focus on zero-coupon bonds. Using a randomly chosen historical date, one can see the term structure of zero-coupon bonds by deducing the rates from government bond prices, which is explained in detail in Chapter 4. For now, let's examine the rates and see what information they contain.

We use the following notation:

Notation	Meaning
r_{0i}	Annualized spot rate for a risk-free zero-coupon bond maturing at time i, and bought at the present time 0. This value is known when the bond is purchased.
d_i	The present value of a riskless $1 paid at time i, either $(1+r_{0i})^{-i}$ for discrete compounding or $\exp[-r_{0i}i]$ for continuous compounding
r_{ij}	With j > i > 0, this means an annualized future spot rate for a zero-coupon bond bought at time i that matures at time j. *This value is unknown at time 0.*
$E[r_{ij}]$	The expected value of r_{ij} based on information at time 0. This may be assumed or based on a model.
f_{ij}	The forward rate of interest from i to j, determined at time 0. This is the annualized rate of return locked in from time i to time j achieved by buying the bond with maturity j now, and earning r_{oi} until the forward bond starts.

To motivate our discussion, suppose we want to know the one year spot rate one year from now. We currently may know the one year spot rate and the two year spot rate, but we do not know what the one year spot rate will be in one year. When the dates are one period apart, we could say (discrete followed by continuous):

$$d_2 = \frac{d_1}{1+r_{12}} \qquad or \qquad d_2 = d_1 e^{-r_{12}}$$

Then, we would realize that r_{12} is unknown to us, usually modeled as a random variable. Is there a constant I can substitute for r_{12} to make this equation true? $E[r_{12}]$ is a tempting answer, but we don't really know the expectation either, unless we assume a form of the random variable. But even if one could assume a probability distribution, there is no guarantee that the value of the expected value would make this equation true.

What if we define a constant, the forward rate, that forces the equation to be true? Then it would be as if we bought a two-year ZCB, earned r_{01} in the first year (a known rate) and a locked-in forward rate "f_{12}" from year 1-2. Then the defined forward value would be (discrete followed by continuous, dates one period apart)

$$f_{12} = \frac{d_1}{d_2} - 1 \qquad or \qquad f_{12} = \ln(\frac{d_1}{d_2})$$

When the dates are not one period apart, and the distance is $t_2 - t_1 > 0$,

$$f_{12} = \frac{d_1}{d_2}^{1/(t_2 - t_1)} - 1 \qquad or \qquad f_{12} = \ln(\frac{d_1}{d_2})/(t_2 - t_1)$$

To summarize, the forward rate is the single fixed rate in year 1-2 that gives us the same return as if we had bought a 1 year ZCB and reinvested the proceeds to receive exactly $1 at time 2. It is not the same as r_{12} (a random variable) or even its expectation. It is the price of money from year 1-2 implied by bond spot prices that replaces the value of the unknown future cost. An investor would be equally happy with accepting an implied fixed return of f_{12} in exchange for the unknown rate r_{12}, as long as they accept the ZCB bond prices. Therefore, the random variable r_{12} (the unknown cost of money from time 1 to 2) may be replaced with f_{12}, which is treated as a constant and discounted at a risk-free rate.

The term f_{12} in this context is called the *risk-neutral* equivalent of the unknown r_{12}. In this context, it can also be called the forward rate, or the forward price of money from time 1 to 2, implying that it is a rate determined now for future time periods.

Risk-neutral values are one of the single most important concepts in finance, since they can be used to replace risky future cash flows having unknowable discount rates with risk-neutral quantities discountable at risk-free rates. Without risk-neutral equivalents, we need to build models to determine the proper costs of risk and make the necessary adjustments. With risk-neutral values, no further risk adjustments need be made.

To illustrate the concept in practice, the table below shows the spot rates of ZCBs at a point in time as a function of different maturities measured in years. The rates are called spot rates because the bonds are purchased and paid for on a spot basis. Therefore, one could invest for 1 year and earn 1.22%, or 2 years and earn 1.40% per year.

Year				Simple
Start	End	Spot	Forw	Avg
0	1	1.22	1.22	1.22
1	2	1.40	1.58	1.40
2	3	1.54	1.83	1.54
3	4	1.67	2.04	1.67
4	5	1.78	2.24	1.78
5	6	1.89	2.41	1.89
6	7	1.98	2.57	1.98
7	8	2.07	2.71	2.08
8	9	2.16	2.84	2.16
9	10	2.24	2.96	2.24

One might also ask, what return is earned from year 1 to year 2? Can we determine the implied return from the given data? You might say, if I make 1.22% for one year (0-1) and 1.40% on average for 2 years (0-2), it is as if I am earning 1.58% in the second year, so that the average rate would be 1.40%. If you said this, you would be approximately correct. If you were discounting continuously, you would be exactly correct! The exact discrete formula would suggest comparing two strategies, specifically a two-year investment, or a one-year investment being guaranteed a given *forward rate* in year 1 for 1 year. These strategies should yield the same result. Therefore,

$$(1 + r_{02})^2 = (1 + r_{01})(1 + f_{12}) \qquad discrete\ formula$$

$$e^{2r_{02}} = e^{r_{01}} e^{f_{12}} \rightarrow 2r_{02} = r_{01} + f_{12} \qquad continuous\ formula$$

The term f is called a forward rate because it implicitly allows us to compute the return as if we had a contract to purchase the 1-year bond in 1 year with the proceeds of a 1-year bond invested in one year. The forward rate curve, we can see, rises faster than the spot rate. Since the spot rate is approximately the average of the forward rates, this makes sense. For the average spot rate to rise over time, the forward price must be increasing faster.

VALUING RISK-FREE CASH FLOWS WITH VARIABLE INTEREST RATES

Suppose we have a sequence of known cash flows of $100 each in years 1-10, continuing the example above. The present value of the cash flows is quite simple in the case, since we can discount each cash flow separately at the corresponding spot rate of interest. The result is $900.77, which is shown in the Excel calculation below, along with a computation of the *Internal Rate of Return* (IRR),a single average compounded yield corresponding to the pricing of the cash flows.

Date	Spot	PV(100)	CF
0			-900.77
1	1.22%	98.79	100
2	1.40%	97.26	100
3	1.54%	95.51	100
4	1.67%	93.60	100
5	1.78%	91.55	100
6	1.89%	89.40	100
7	1.98%	87.16	100
8	2.07%	84.85	100
9	2.16%	82.51	100
10	2.24%	80.14	100
	Total	900.77	
	IRR		1.95%

Another application associated with variable interest rates has to do with cash flows that change as a function of the interest rates. In particular, a *floating rate bond or note* is one whose coupons vary based on a pre-specified interest rate index. How should we value floating rate bonds?

First, we should know that the interest rate on a floating rate bond is normally determined at the beginning of the payment period, not the end. This convention of declaring the interest rate at the beginning of the effective period is known as quoting interest *in arrears*. For example, a floating rate bond with annual payments would be structured so that the first payment equals the one-year spot rate now at time 0 (r_{01}) multiplied by the principal balance (F). Using the GVE, we can write it this way:

Valuation of a one-period FRN

$$r_{01}V_0 = (F - V_0) + r_{01}F$$

This implies that the floating rate bond is priced at par, or $V_0 = F$. If we had a two-year floating rate bond, the price at time 1 and 0 would be determined by

Recursive valuation of a two-period FRN

$$At\ time\ 1: \quad r_{12}V_1 = (F - V_1) + r_{12}F \ \ or \ \ V_1 = F$$

$$At\ time\ 0: \quad r_{01}V_0 = (V_1 - V_0) + r_{01}F \ \ or \ \ V_0 = F$$

implying that after each floating coupon is paid, the value of the bond resets to the principal value, *regardless of the value of the new spot rate*. This is important especially since r_{12} is not known at time 0. The value of floating rate bonds constructed in this matter is straightforward --- they always stay at par value if they are risk-free, at the moment the last payment is made. Interest may accrue, i.e. become owed but not included in the bond price, but the bond price resets to par after the floating interest rate is paid.

What if we needed to value a single cash flow that was a function of the future spot interest rate, and not part of a floating rate bond? In this case, the unknown r_1 would not cancel as neatly as with the FRN. This would occur for example if you were trying to value a company whose earnings fluctuated with interest rates. In this case, we would need to find a number that represented the market value of the variable payment. Fortunately, we already know the forward rate serves that role, and that information is embedded in the spot yields. Therefore, if the actual cash flow equals $r_{t-1,t}\,C(t)$ at time t, the present value is given by

$$PV = \frac{f_{t-1,t}\, C(t)}{(1 + r_{0t})^t}$$

The forward rate f in this formula represents the forward rate one year prior for a payment on the same payment date. Continuing the previous example, suppose we want to value each FRN bond payment separately. We substitute the forward price for the unknown future spot price and discount it at the risk-free spot rate. Here are the details:

FRN Payments - Individual Present Value:

F=100		Forw	PV using
	Spot	Equiv	Spot
1	1.22%	1.22	1.21
2	1.40%	1.58	1.54
3	1.54%	1.83	1.75
4	1.67%	2.04	1.91
5	1.78%	2.24	2.05
6	1.89%	2.41	2.16
7	1.98%	2.57	2.24
8	2.07%	2.71	2.30
9	2.16%	2.84	2.34
10	2.24%	102.96	82.51
		Total	100.00

Of course, the total value is *F = 100*, as expected. This demonstration shows us how to value any risk-sensitive cash flow at any time to a risk-free forward equivalent, and determine its value by discounting at the risk-free spot rate.

Example

A company has determined that its cash flows for the next three years are sensitive to interest rates. The estimated cash flow each year is estimated to be $100 MM times the one-year spot rate in the previous year. What is the present value of the first 3 cash flows using the same term structure as the one shown in this section?

Answer: $4.5 mm.

3.6 FOREIGN EXCHANGE

The U.S. Federal Reserve Bank reports spot exchange rates for the major currencies on every trading day. "Spot rates" refer to the exchange rate on a given day, while "Forward rates" refer to exchange rates for future transactions whose terms are agreed today. In particular, the exchange rate to be applied in the future is determined today. According to Investopedia, the interbank foreign exchange market trades on average about $5 trillion *per day* in 2023, making it the largest capital market in the world.

Most currencies are quoted in terms of the number of units of foreign currency received for each $1 USD. This reporting standard is called a *direct quote*. There are a few exceptions, as noted with asterisks (*) in the following table (from 12/22/2017):

COUNTRY	CURRENCY	CODE	12/22/2017
*AUSTRALIA	DOLLAR	AUD	0.7713
BRAZIL	REAL	BRL	3.3209
CANADA	DOLLAR	CAD	1.2748
CHINA, P.R.	YUAN	CNY	6.575
DENMARK	KRONE	DKK	6.2866
*EMU MEMBERS	EURO	EUR	1.1839
HONG KONG	DOLLAR	HKD	7.8137
INDIA	RUPEE	INR	64.02
JAPAN	YEN	JPY	113.3
MALAYSIA	RINGGIT	MYR	4.077
MEXICO	PESO	MXP	19.5905
*NEW ZEALAND	DOLLAR	NZD	0.7015
NORWAY	KRONE	NOK	8.3387
SINGAPORE	DOLLAR	SGD	1.3441
SOUTH AFRICA	RAND	ZAR	12.605
SOUTH KOREA	WON	KRW	1076.87
SRI LANKA	RUPEE	LKR	152.85
SWEDEN	KRONA	SEK	8.358
SWITZERLAND	FRANC	CHF	0.9898
TAIWAN	DOLLAR	TWD	29.89
THAILAND	BAHT	THB	32.73
*UNITED KINGDOM	POUND	GBP	1.3379
VENEZUELA	BOLIVAR	VEF	9.975

*indirect quotes: USD per currency unit

Near the end of 2017, the spot exchange rate for Chinese Yuan (CNY) was 6.50 Yuan per $1 U.S. dollar (USD).



We can see that the spot CNY exchange rate grew from about 6 to about 7 in three years 2014-2017, implying an annual growth rate of about 5%. A positive growth rate indicates that the CNY was weakening relative to the dollar, since more units of the Chinese currency were required to purchase 1 USD as time passed. We might like to know how we should expect to see the exchange rate move in the future.

There are many theories relating to the determination of an exchange rate between two currencies, but the main drivers are relative interest rates, inflation, current account (trade) deficits, public debt levels, political

stability and economic performance[8]. The best available indicator of expected future exchange rates is the *forward* exchange rate, ignoring any risk premium. A forward contract may specify for example that $1 could be exchanged for CNY 7 in 5 years time. This would indicate expected future weakening of the currency.

There is a strict relationship between spot rates, forward rates and interest rates governing relative prices known as IRP or **Interest Rate Parity**. The table below demonstrates IRP by showing two ways to invest 1 USD to produce CNY in the future. It can either be invested in the U.S. and converted to CNY using a forward contract, or it can be converted now at the spot exchange rate and invested in a Chinese bond.

Interest rate parity illustration for direct quotation currencies

Time	Now	In t years	In CNY
USD	$1	$(1+r_{USD})^t$	$F(t)(1+r_{USD})^t$
CNY	¥1 (X_{CNY})	$X_{CNY}(1+r_{CNY})^t$	$X_{CNY}(1+r_{CNY})^t$

The final column entries are guaranteed to be equal by the threat of arbitrage; any discrepancy can be made profitable by borrowing in one currency to invest in the other, until the discrepancy disappears. This implies that F(t) forward prices at time t are completely determined by the spot exchange rate (X) and the ZCB interest rates of the two currencies.

The interest rate parity equation is a restatement of the equality of the last two table entries:

$$\frac{F(t)}{X_{CNY}} = \frac{(1+r_{CNY})^t}{(1+r_{USD})^t} \equiv (1+\delta)^t$$

If the Chinese interest rate exceeds the US interest rate, then the USD will become more expensive to purchase over time at the rate of δ, the CNY currency depreciation rate.

In sum, the purpose of this section is to allow you to model foreign-currency denominated cash flows. Rather than trying to estimate the expected future exchange rate, one can convert future foreign payments into domestic payments at the forward rate and then discount them at the risk-free rate. Alternatively, foreign risk-free payments can be discounted at the foreign interest rate and converted to domestic currency using the spot exchange rate.

[8] There are also instances where the spot exchange rate is fixed or pegged. One famous example is Argentina's pegging of the peso to the USD from 1991 to 2002.

3.7 SUMMARY

This chapter may look like a seemingly unconnected list of useful cash flow tricks, but it is more than that. Our objective by the end of Part I is to be able to simulate and value cash flows of any type. The purpose of this chapter was to introduce you to the many alternatives. For example, your cash flows might be subject to timing irregularities, different compounding frequencies, and inflation adjustments. They might be subject to taxes or event risks.

In other cases, we might have unknown variables affecting our valuations, such as changing interest rates and inflation rates. While we have not yet studied models of stochastic rates, we did learn how to recover forward interest rates and forward foreign exchange rates from underlying ZCB values and spot exchange rates. As it turns out these forward rates take the place of the unknown variable rates in the future for valuation purposes. When these forward rates are included in cash flow determination, the cash flows can be discounted at risk-free rates.

Chapter 3 was your introduction to risk-neutral valuation, a topic that will remain important throughout this text.

3.8 REFERENCES AND FURTHER READING

3.9 EXERCISES FOR THE CALCULATOR

1. Which has a higher effective annual yield, 10% interest compounded quarterly or 9.8% compounded continuously?

2. *(Harder than it looks!)* Consumer home loans (mortgages) usually have monthly payments. Some companies are advertising the possibility of making 26 biweekly payments instead of 12 monthly payments. If the simple annual rate for a normal mortgage is 4.5%, and we have a $500,000 loan for 15 years, what are the payments on the regular mortgage and on the biweekly mortgage? What is the sum of all the payments in both cases? Which do you think is better? *Note: The biweekly mortgage payment is computed as half of the regular monthly payment, meaning that the mortgagor would make 13 monthly payments a year.*

3. Rank the following mortgage offers in terms of which is best for the consumer, using a 30 year loan of $225,000. Be sure to list your assumptions and explain your reasoning.

    a. A 3.5% mortgage with 3 points in closing costs
    b. A 3.75% mortgage with 1.5 points in closing costs
    c. A 4.00% mortgage with no closing costs

    Note: A loan with 3 points in closing costs implies that the consumer must pay a fee of 3% of the value of the mortgage as compensation to the bank or broker. The fee is normally financed, meaning that the payment is computed using the loan amount plus the closing costs as the principal

4. If the nominal interest rate is 5% and the inflation rate is 2% (simple annual rates), then what is the real interest rate? What is the answer if the two rates are continuously compounded rates?

5. Your employer gives you a choice of two compensation plans. In Plan A, your base wage will be increased now by 10% and stay flat for the next 10 years. In Plan B, your current wage will be indexed to inflation. The nominal interest rate is 5%, and the expected inflation rate is 2%. What is the present value of each plan? What other factors might cause you to choose Plan A over Plan B?

6. Your pension plan offers you one of two options:

    a. A fixed annuity of $5000 per month
    b. An inflation-indexed annuity starting at a $4500 base level today, but with the first payment in one month increasing at the rate of inflation.

    If the interest rate is 5% and the inflation rate is 2%, what is the value of each plan, assuming you are going to live 20 years longer?

    If you have a 0.25% chance of dying per month, what are the new plan values? HINT: Divide all annual rates by 12 to determine monthly rates.

7. A stock is expected to pay 2% of its value in dividends per year, and grow at a net rate (after dividends) of 7% per year. If the capital gains tax rate is 15% and the ordinary income tax rate is 40%, then what is the expected after-tax return on the stock? Note: Dividends are taxed as ordinary income.

8. A bank's energy loan portfolio is estimated to have an average LGD of 70% given that oil prices have halved in the last few years. If the default probabilities are 8% per year, the principal is $100,000,000, and the scheduled interest payments are fixed at 6% per year for 12 years, what is the value of the energy loan portfolio today? How does your answer change if oil prices fall further? Assume the risk-free rate is 4%. *Note:* LGD (loss given default) is the percentage loss of each promised payment in the event of default.

9. If you bought a single payment life insurance policy for $150,000 today which paid $2,000,000 on death, the risk-free interest rate is 4%, and there is no risk premium, what is the implied annual probability of death?

3.10 EXERCISES TO BE DONE IN EXCEL

A. Revise the Excel tax computation in the chapter to value a retirement annuity instead of a bond. The annuity pays $10,000 per year for 20 years, and the cost of the annuity is determined by discounting the payments at the contract rate of 3% per year.

    1. What is the valuation and effective after tax rate for the annuity if the capital gains rate is 15%, the ordinary income rate is 40%, and the market risk-free discount rate is 4%? Assume the annuity is purchased from pretax funds. What is the value from the annuity provider's point of view?

    2. What is the answer to part 1 if the annuity is purchased from after tax funds?

NOTE: Annuity taxation differs from bonds. This quote is from Kiplinger in the context of retirement annuities: "If you buy the annuity with pretax money, then the entire balance will be taxable. If you use after-tax funds, however, then you'll be taxed only on the earnings. If you cash out a deferred annuity in a lump sum, then you'll have to pay income taxes on all of the earnings higher than your original investment. Jul 10, 2009".

MINIPROJECT 3: BOOTSTRAPPING THE U.S. TREASURY ZCB YIELD CURVE FROM BOND PRICES

Ask your instructor to choose a more recent date or download U.S. Treasury Notes and Bond quotations as of 2/8/23 from the Wall Street Journal. You are about to perform one of the most important functions of an investment bank, which is to know the exact ZCB yield curve at any point in time. The risk-free ZCB curve is used to determine forward interest rates, foreign exchange forward rates and derivative prices. Any cash flow receipts or commitments of the bank must be continually revalued as the term structure changes. Therefore, you should consider this problem seriously.

The details of bootstrapping are found in section 4.4. The idea is to imagine that we are trying to find the continuous discount factor $d(t)=\exp(-r(t)t)$ for all dates t from 0 to 30 years. The term d(t) is a complex and unparameterized function of time. We will compare the pricing of alternative d(t) parameter values to find the parameter set that finds the closest fit between our model prices and the actual market prices of all the bonds.

You report should include the following:

1. Your completed Excel spreadsheet
2. A plot of r(t)
3. A plot of market price vs theoretical price

Chapter 4: Bonds

INTRODUCTION

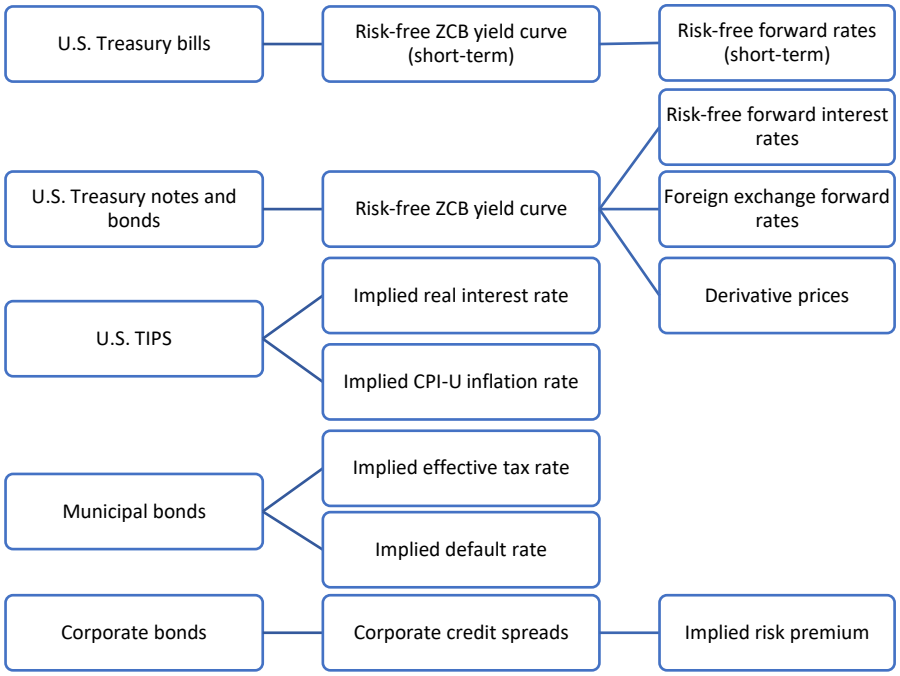
From one perspective, any financial engineer expecting to work in a capital markets function must learn the operational details of bonds and bond valuation. Once an analyst knows the yield to maturity, the coupon accrual conventions, the quoting conventions, applicable tax treatment, inflation rate, default rates, recovery rates and everything else that affects the value of the bond, he or she may proceed to value the bond with some confidence.

Although valuation is important, market price is usually the better measure of value than the result of a theoretical model. Hence, together with a model, bond prices can also be used to estimate the *implied* model parameters that might be unknown or difficult to estimate. In Chapter 3, for example, we learned how to discover bond forward rates and currency forward prices from the observed ZCB rates. In this chapter, we will extend this logic to explore the determination of the 30-year implied ZCB yield curve, real rates of interest, inflation rates, tax rates, default rates and estimates of loss given default (LGD) for corporate bonds.

Ultimately, we are focused on valuation of all cash flows, and therefore the information we collect from bond prices informs our valuations. Therefore, our purpose on the chapter is threefold:

- Understand and build models to value fixed income assets
- Discover implied valuation model parameters from fixed income asset and contract prices
- Use this information to value other cash flows

The chart below summarizes much of the information that can be extracted from bond prices.

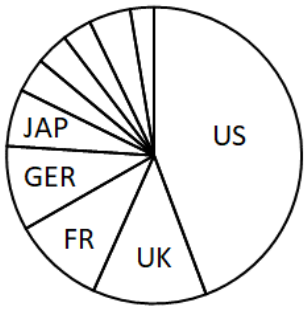


4.1 GLOBAL AND U.S. GOVERNMENT DEBT

The top 10 countries in the world had about $70 trillion in external debt at the end of 2022. The largest debtor in the world was the U.S. at that time, with about $31 trillion outstanding.

External Debt - Top 10 Countries - 2022

	USD trillions	%
United States	31	44%
United Kingdom	8.73	12%
France	7.04	10%
Germany	6.46	9%
Japan	4.36	6%
China	2.64	4%
Italy	2.51	4%
Spain	2.26	3%
Canada	3.2	5%
Australia	1.83	3%
Total	70.03	100%



Source: Wikipedia: List_of_countries_by_external_debt

The U.S. $31 trillion figure includes marketable debt and non-marketable debt. Marketable debt is in the form of tradeable U.S. government bonds, which collectively represent the largest fixed income market in the world. Marketable U.S. government bonds were worth about $24 trillion in 2022, with the remaining $7 trillion of debt being composed of "Federal employee & veteran benefits payable" and "other."

The federal debt does not include the following (2022 figures for 1 & 2 from SIFMA)[9]:

1. State and local (municipal) debt, $4 trillion
2. U.S. government agency debt (e.g. FHLB, FNMA, FHLMC, FFCB, …), $1.9 trillion
3. Unfunded liabilities of entitlement programs (e.g. Social Security and Medicare), $7.5 trillion?

Most U.S. government debt is sold in the form of tradeable securities. However, there are exceptions. Savings Bonds, Series EE and Series I, are held electronically and owned by individuals, unable to be traded. U.S. Treasury securities do not trade on any exchange. They are initially sold at auction to bond dealers, but are then traded *over-the-counter*, that is, privately between counterparties. TreasuryDirect.gov acts as a counterparty for retail and other investors, and is the exclusive manager of U.S. Savings Bonds.

4.2 U.S. GOVERNMENT SECURITIES

Marketable securities include Treasury Bills (T-Bills), STRIPs, T-Notes, T-Bonds, Treasury Inflation Protected Securities (TIPS), and Floating Rate Notes (FRNs). These instruments are considered risk-free in the sense that the U.S. government is highly unlikely to default on the debt. This is because the debt is denominated in dollars, and the issuance of dollars is controlled by the U.S. government. Hence, while investors are guaranteed to receive dollars, they are not guaranteed what those dollars will be worth. Also, while the payments are considered risk-free, bond prices will change over time as interest rates change, causing losses for investors who sell those bonds

[9] Entitlement program shortfalls are normally reported as annual projections, subject to actuarial assumptions. Present value figures are hard to find. However, cumulative 10-year projected deficits for the two programs are estimated at $2.5 trillion and $5 trillion.

prior to maturity. Therefore, the term risk-free should be considered to apply only to the level of the promised payments --- not to the market value of bonds.

A brief history

U.S. debt dates back to the Revolutionary War in 1776, but at that time, the debt was privately held by individuals and foreign governments. T-notes were introduced in 1812 and were originally issued as a means to finance the war with Britain. T-notes offer semiannual fixed payments and a range of 2-10 years to maturity from the issuance date. Principal is repaid on the maturity date. T-bonds are identical in structure to T-notes, except that the maturity at issuance runs from 10 to 30 years.

T-Bills were first introduced in 1929, in order to finance U.S. government programs during the Great Depression. The auction strategy for selling T-bills allowed more flexibility and higher frequency of issuance than existed for bonds and notes. The denomination of each T-Bill is $10,000, payable as a single principal payment due within one year of issuance. T-Bills are classified as short-term ZCBs (zero coupon bonds) because they are sold at a discount to par and do not make any payments prior to maturity. The short-term designation refers to the maturity occurring with one year of issuance.

The U.S. Treasury STRIPS (Separate Trading of Registered Interest and Principal of Securities) program was first introduced in February 1985. The program allows investors to hold and trade individual interest and principal payments of eligible Treasury notes and bonds as separate securities. In other words, STRIPS convert coupon bonds into portfolios of ZCBs. Since STRIPS maturities range from 0-30 years, they are an excellent reference point to estimate the risk-free discount rate for any given date within that range. STRIPS quotes were once widely reported but are no longer readily available. For this reason, we do not focus much on STRIPs in this text. Also, most analysts prefer to use T-note and T-Bond prices to determine the ZCB yield curve, given that the bond market is much deeper and is more actively traded, providing more reliable pricing.

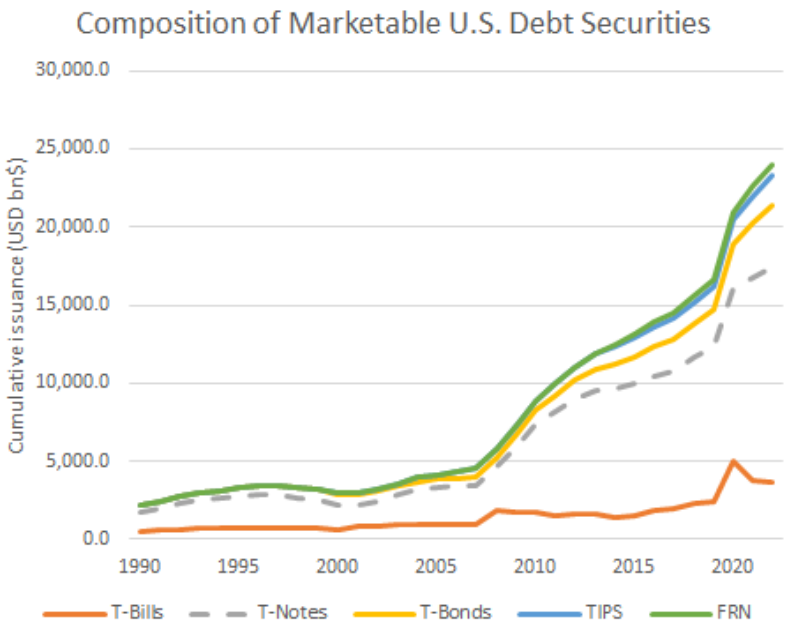
TIPS were created in 1997, and then expanded from a 20-year maximum maturity to 30 years in 2009. These securities provide investors with interest and principal payments that adjust according to the CPI-U inflation rate, defined later in the chapter. Floating rate notes (FRNs) were introduced in 2013. The floating rate payment is based on the most recent 13-week T-Bill auction high rate, and FRNs are issued at 2-year maturities.

Taxation of U.S. Government bonds from the Raymond James website

“Interest income from Treasury securities is subject to federal income tax but exempt from state and local taxes. Income from Treasury bills is paid at maturity and, thus, tax-reportable in the year in which it is received. Although not paid until maturity, income from zero-coupon STRIPS is taxable in the year in which it accrues. Increases in TIPS’ principal value as a result of inflation adjustments are taxed as capital gains in the year they occur, even though an investor does not collect these gains until TIPS are sold or mature. This is known as a ‘phantom income’ tax. Conversely, decreases in the principal amount due to deflation can be used to offset taxable interest income from other investments.”

Market size for U.S. Treasury issues

The sizes of the five marketable security components of the government debt are shown in the table below, from 1990-2022. The stacked line chart shows the cumulative amount of financing, with the bottom line representing T-Bills outstanding, with each successive category being added to the previous sum.



U.S. Marketable Debt as of Dec 31, 2022*	
Bills	3,697
Notes	13,752
Bonds	3,960
TIPS	1,908
FRN	617
Total	23,934

*USD Billions

Source: SIFMA data, author's graph

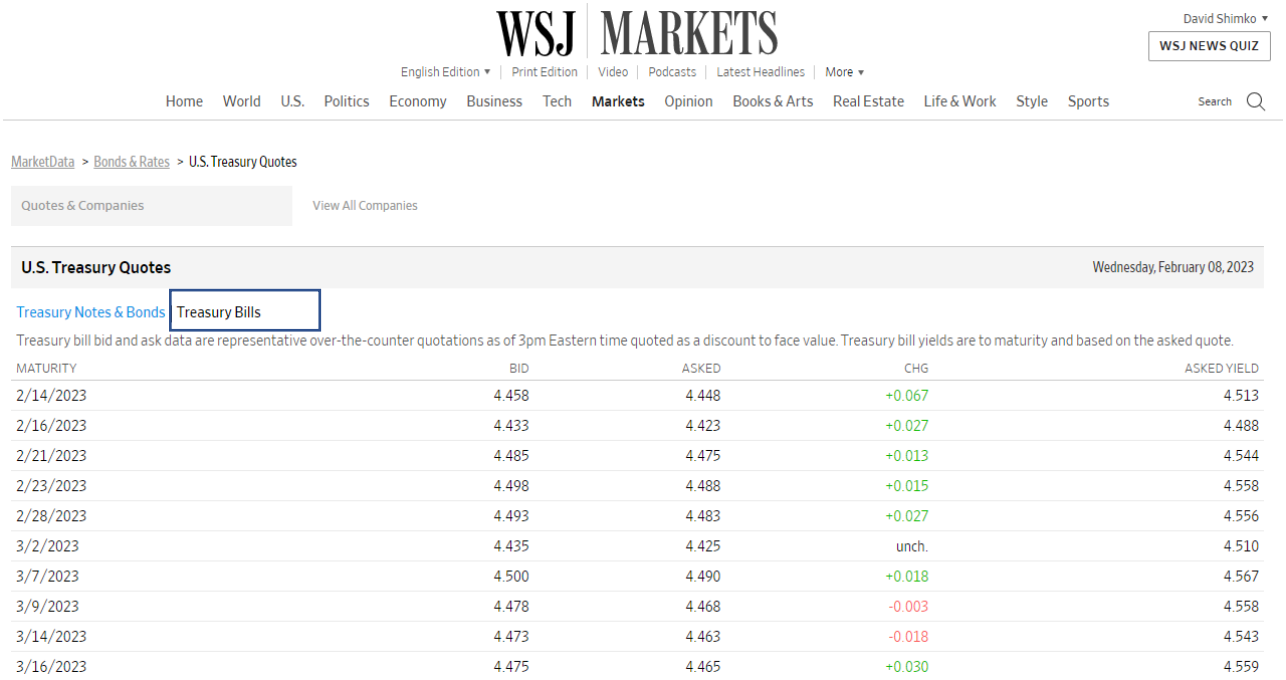
In terms of issuance, T-notes historically represented the largest single component with 2-10 years maturity at issuance. Note that STRIPs are stripped interest and principal payments from notes or bonds, and therefore are included in the notes or bond categories.

Marketable debt is unsustainably increasing at an increasing rate, the result of increased spending in both the Trump and Biden administrations, fueled in part by COVID-related expenditures, investment losses, reduced tax collections, increased entitlements, and the war in Ukraine. As interest rates rise, the cost of servicing the new debt is also increasing rapidly. Challenging economic conditions have important implications for bond markets and the risks associated with bond trading and bond portfolios generally. For these reasons, the financial engineer must be well-grounded in the valuation and risk assessment of fixed income investments.

4.3 TREASURY BILLS

With less than one year to maturity, and no coupons, T-Bills are an ideal ZCB benchmark for valuing short-term cash flows. They are issued by the U.S. government and therefore are considered risk free. The face value at maturity is $10,000 per T-Bill.

Here is a WSJ quote excerpt from 2/8/23:



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U.S. Treasury Quotes Wednesday, February 08, 2023

Treasury Notes & Bonds | Treasury Bills

Treasury bill bid and ask data are representative over-the-counter quotations as of 3pm Eastern time quoted as a discount to face value. Treasury bill yields are to maturity and based on the asked quote.

MATURITY	BID	ASKED	CHG	ASKED YIELD
2/14/2023	4.458	4.448	+0.067	4.513
2/16/2023	4.433	4.423	+0.027	4.488
2/21/2023	4.485	4.475	+0.013	4.544
2/23/2023	4.498	4.488	+0.015	4.558
2/28/2023	4.493	4.483	+0.027	4.556
3/2/2023	4.435	4.425	unch.	4.510
3/7/2023	4.500	4.490	+0.018	4.567
3/9/2023	4.478	4.468	-0.003	4.558
3/14/2023	4.473	4.463	-0.018	4.543
3/16/2023	4.475	4.465	+0.030	4.559

The legacy quotation system for trading and reporting T-Bill rates presents a minor issue. This can easily be resolved once we know the conversion from a rate quote into a price quote:

$$T-Bill\ Asking\ Price = 10{,}000\left(1 - Quoted\ asked\ \times \frac{Days}{360}\right)$$

From the price, we can compute the continuously compounded yield using:

$$r = \frac{-\ln\left(\frac{Price}{10{,}000}\right)}{\left(\frac{Days}{365}\right)}$$

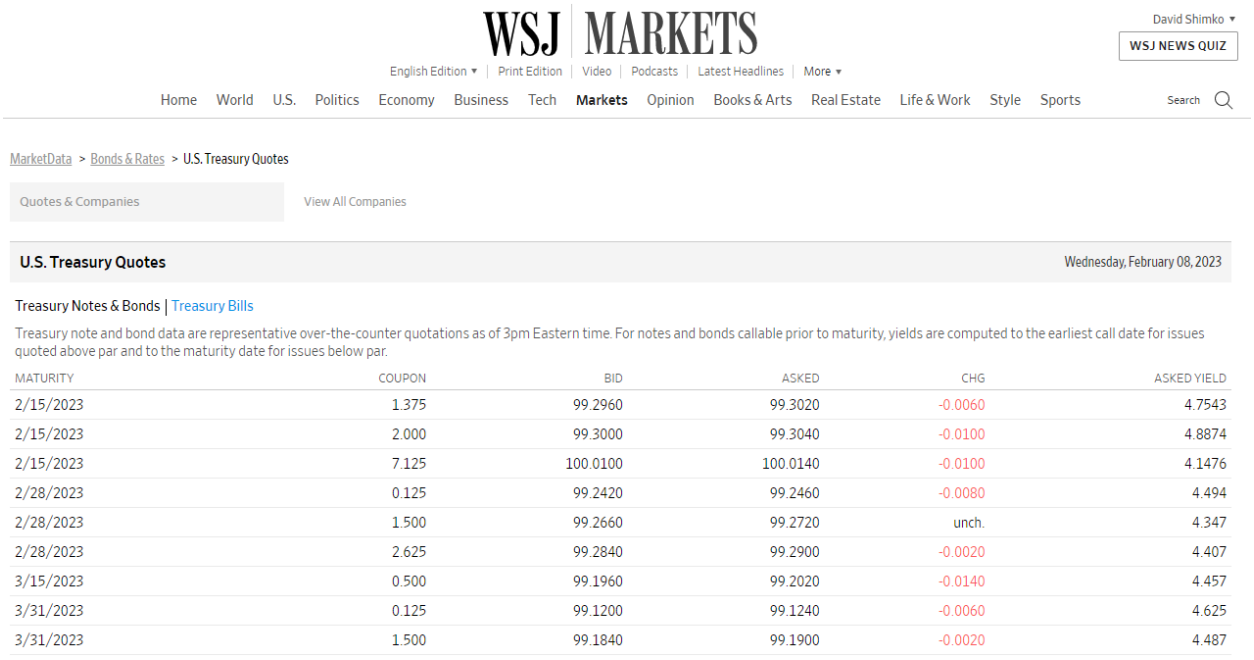
For example, the 3/16/23 T-Bill maturity (36 days) shows an asking quote of 4.465 percent. This means the price is $10{,}000(1 - 0.04465\left(\frac{36}{360}\right))$=\$9,955.35. The continuously compounded yield is 4.54% p.a., quite close to the reported asked yield of 4.559%.

Please note: The WSJ yield calculation differs slightly from ours. Please see https://www.treasurydirect.gov/instit/annceresult/press/preanre/2004/ofcalc6decbill.pdf for complete details.

4.4 U.S. GOVERNMENT COUPON BONDS – T-NOTES AND T-BONDS

U.S. government notes and bonds make coupon payments semiannually. Their quoted face value is \$100, and the semiannual coupon is given by 100c/2, where c is the annual coupon rate, so a 6% coupon bond pays \$3 every six months until the bond matures. On the maturity date, the bond pays the final coupon and the \$100 principal.

Here is a sampling from the Wall Street Journal's reported bond prices on 2/8/23. Altogether, there were 328 bond quotes posted that day for different maturities.



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U.S. Treasury Quotes Wednesday, February 08, 2023

Treasury Notes & Bonds | Treasury Bills

Treasury note and bond data are representative over-the-counter quotations as of 3pm Eastern time. For notes and bonds callable prior to maturity, yields are computed to the earliest call date for issues quoted above par and to the maturity date for issues below par.

MATURITY	COUPON	BID	ASKED	CHG	ASKED YIELD
2/15/2023	1.375	99.2960	99.3020	-0.0060	4.7543
2/15/2023	2.000	99.3000	99.3040	-0.0100	4.8874
2/15/2023	7.125	100.0100	100.0140	-0.0100	4.1476
2/28/2023	0.125	99.2420	99.2460	-0.0080	4.494
2/28/2023	1.500	99.2660	99.2720	unch.	4.347
2/28/2023	2.625	99.2840	99.2900	-0.0020	4.407
3/15/2023	0.500	99.1960	99.2020	-0.0140	4.457
3/31/2023	0.125	99.1200	99.1240	-0.0060	4.625
3/31/2023	1.500	99.1840	99.1900	-0.0020	4.487

The bid and ask prices are quoted here in decimal form, although prices used to be reported in 32^{nd}s. The *asked yield* equals 2 times the semiannual yield to maturity of the bond, and is also called a *bond equivalent yield*. The yield calculation is based on the asking price, since this is the cost of a bond to the bond buyer.

The bid and ask prices are known as *clean prices*, since they do not include *accrued interest*. For example, the bond maturing 2/15/23 paid its previous coupon six months before the expiration date, i.e. 8/15/22. Since that time, 177 days have *actually* passed out of 184 *actual* days in the coupon period. This is called the ACTUAL/ACTUAL convention for computing accrued interest accrued interest in the U.S. bond market. To compute the accrued interest, we take $177/184^{ths}$ of the semiannual coupon payment (1.375/2) to get the dollar amount, which is \$0.6613. Therefore, if you were to buy the bond, you would pay the *invoice price*, which would be the asking price of 99.3020 plus the accrued interest of 0.6613, for a total of \$99.9633. The total cost is also sometimes called the *dirty price* of the bond.

Bootstrapping the Yield Curve

We would like to use the pricing information from coupon-paying bonds to estimate the correct discount rate for ZCBs maturing any date in the future. This process is called *bootstrapping* the yield curve. There are many techniques to bootstrap, but we will only present one here.[10] Some methods rely on advanced interpolation and smoothness maximization techniques. Our method proposes a flexible functional form for the continuously compounded discount rate as a function of time. The function parameters are chosen so

[10] Many yield curve models are based on interpolation rather than nonlinear regression. Interpolation has the benefit of matching market reported prices exactly, but at a cost of functional complexity. The nonlinear regression approach is appropriate if the analyst wants smooth curves and is willing to assume that not all reported prices equal exact "market prices."

that the model's bond prices best approximate market prices. The best parameters minimize the sum of squared error (SSE), i.e. differences between market prices and model prices. This method can be referred to as a nonlinear regression method.

Functional Form Procedure for Bootstrapping

1. Select a function r(t) with undetermined parameters that is sufficiently flexible to provide a wide variety of ZCB term structures, for example

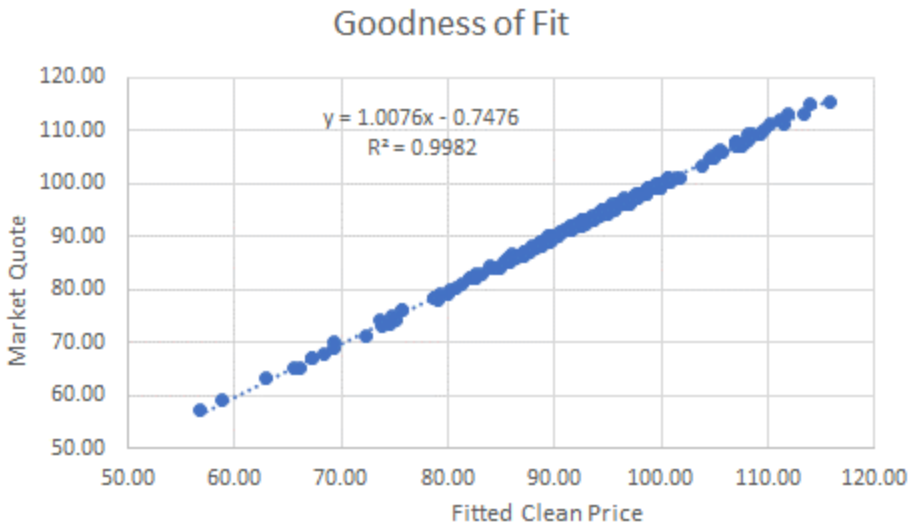
$$r(t) = a + bt + cc\ t^2 + d\ e^{ft} + g\ \ln(t - h)$$

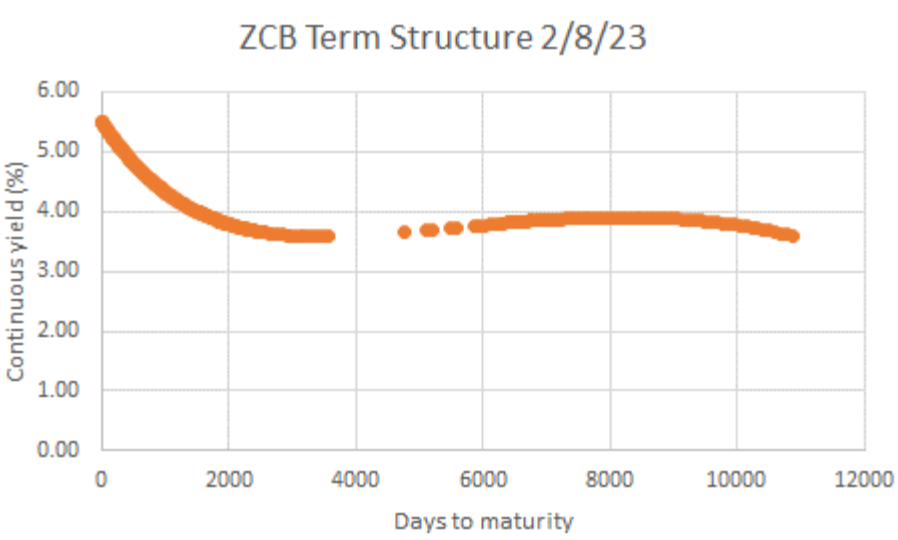
2. For each reported bond, determine the list of coupon dates
3. For each coupon date, determine the number of years (t) between the quotation date and the coupon date. Do the same for the maturity date (T).
4. For each coupon date, compute a discount factor $d_t = \exp[-r(t)t]$. For the principal payment, set $d_T = \exp[-r(T)T]$.
5. For each bond, compute the fitted bond dirty value and subtract accrued interest. The summation is taken over all coupon dates including the maturity date.

$$B = 100 d_T + \frac{c}{2}\sum_t d_t - accrued\ interest$$

6. Compute the sum of squared deviations between the fitted bond price and the market price (most analysts use the average of the bid and ask price as the market price)
7. Use a nonlinear solver (such as in Excel or Python) to minimize the sum of squared deviations between the fitted prices and the market prices by changing the parameters, i.e. {a, b, cc, d, f, g, h}

We leave this as an extremely important and highly recommended task for the reader to bootstrap their own ZCB yield curve. Here are the results we obtained for 2/8/23. The first graph shows the fit between predicted bond prices and the average of the bid and offer prices. The second shows the implied ZCB yield curve, i.e. r(t), using the implied parameters.





a	0.2720
b	0.0086
cc	-0.0001
d	0.0375
f	0.0150
g	-0.1225
h	-7.9728
SSQ	48.83

The parameters that provided the ZCB term structure are here for your reference, to compare with your results. You may like to experiment with different functional forms to see if you can reduce the SSE even further.

Bottom line, we now have a discount factor d_t for every possible date "t" in the future, based on our bootstrapping procedure. This method should not be extrapolated beyond 30 years. However, it can be very useful for valuing risk-free cash flows at time-dependent rates for up to 30 years.

The shape of the term structure

What determines the shape of the term structure? Forward rates are determined by two factors. First, the market has implicit expectations of future interest rates. Expectations of rising rates would be sufficient to produce an upward-sloping yield curve. However, even when the market expects rates to fall, we often see upward-sloping yield curves. Why is this the case?

If all interest rates were the same, bond investors would prefer to lend short-term rather than long-term. However, governments and corporations would prefer to borrow long-term than borrow short-term. Because of this natural imbalance in preferences, long-term rates generally exceed short-term rates to induce lenders (investors) to buy long term bonds. Of course, lower short-term rates give an incentive for corporations to borrow shorter-term. Therefore, there is a *term premium* in equilibrium, i.e. an additional component of the interest rate necessary to clear the risk-free capital markets. We usually write it in the form

$$Forward\ rate = Expected\ future\ spot\ rate + Term\ premium$$

This equation is implied by the *liquidity premium hypothesis*, or can be seen as the likely implication of the *preferred habitat theory*. In the *expectations hypothesis* of the term structure, the term premium is zero. However, few economists would support the expectations hypothesis any longer.

Explaining the ZCB term structure on 2/8/23

You may be wondering why there is a gap in the term structure, and why it takes the shape it does. The gap is caused by a temporary suspension of the issuance of 30-year bonds during the Clinton administration. Because of that suspension, and the fact that more short-term T-notes are issued than longer term T-bonds, there are some dates for which we do not have coupon payments.

But how do we explain the *inverted* yield curve we see in the chart? Suppose we use the equation above to help understand it. If the liquidity premium is positive, then the only way the yield curve can slope downwards is if people expect interest rates will fall in the future by more than the liquidity premium. Since the interest rate is the cost of money, this suggests that the cost of money will fall in the future, possibly due to lack of demand for money for investment. Lack of investment leads to lack of growth.

Of course, the argument is not foolproof. If the market expected high inflation simultaneously, the nominal rate would be higher, possibly leading to an upward-sloping nominal yield curve. It seems that as of 2/8/23, the market is expecting inflationary fears to subside and investment to fall. Time will tell!

4.5 TIPS AND INFLATION

"TIPS" stands for *Treasury Inflation-Protected Securities*. These securities were first issued by the U.S. government in 1997 partly in response to demand for long-term inflation-protected assets by pension funds and long-term investors.

From Treasurydirect.gov:

> "Treasury Inflation-Protected Securities, or TIPS, provide protection against inflation. The principal of a TIPS increases with inflation and decreases with deflation, as measured by the Consumer Price Index. When a TIPS matures, you are paid the adjusted principal or original principal, whichever is greater.

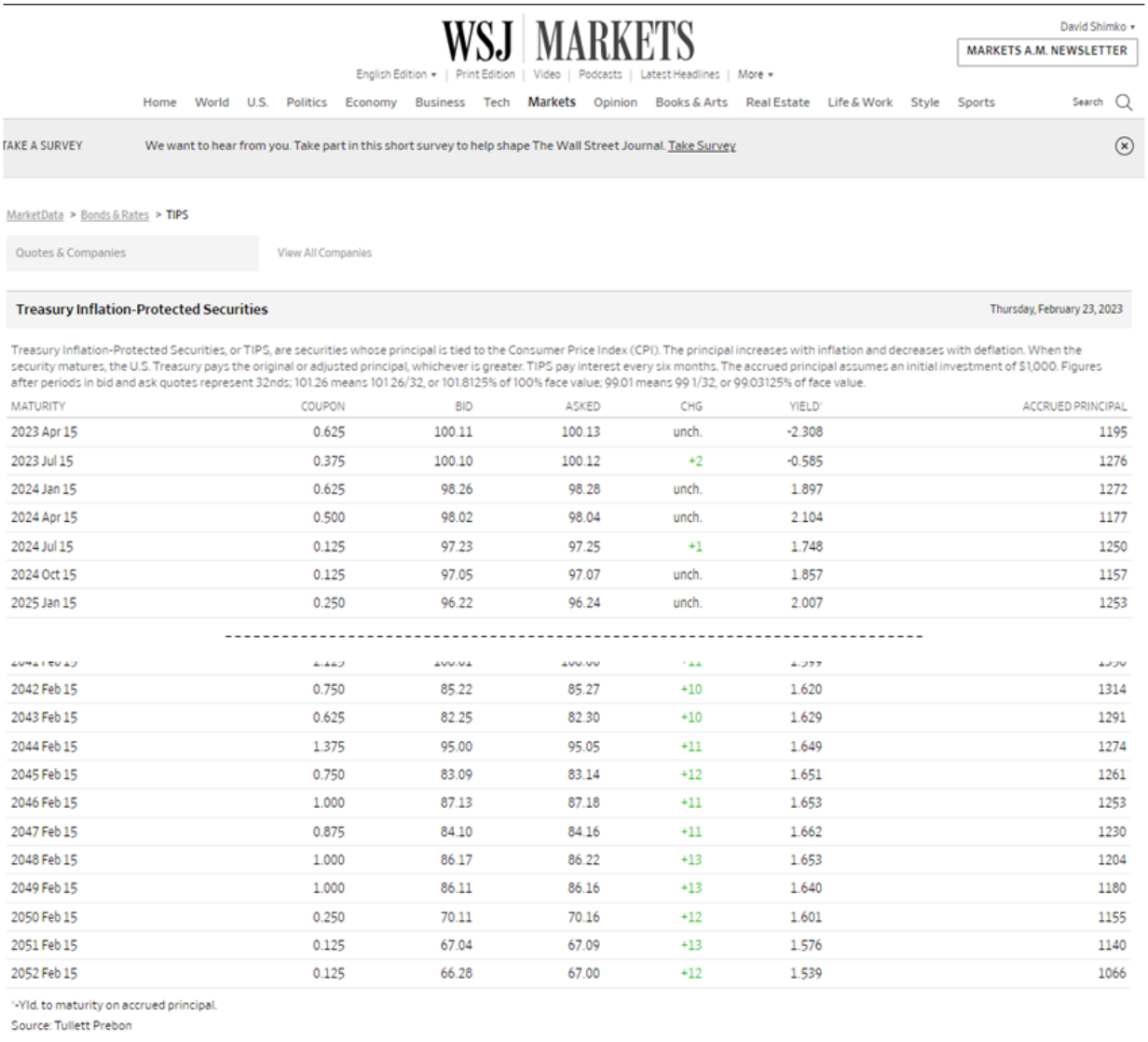
> "TIPS pay interest twice a year, at a fixed rate. The rate is applied to the adjusted principal; so, like the principal, interest payments rise with inflation and fall with deflation."

Investors who purchase TIPS can expect their coupons and principal to appreciate at the same rate as the CPI-U index. However, the tax code inserts a slight wedge in the inflation hedge:

> "Interest payments from Treasury Inflation-Protected Securities (TIPS), and increases in the principal of TIPS, are subject to federal tax, but exempt from state and local income taxes."

Whereas the return of principal in a regular bond is untaxed, the return of inflation-adjusted principal above the face value is taxed. Therefore, this bond is not a complete inflation hedge. However, all is not lost --- the principal also has an embedded option since it pays the higher of the original principal amount or the inflation adjusted principal. This protects against the seemingly remote risk of deflation.

The WSJ report for TIPS quotes on February 23, 2023, showed the following table.



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Treasury Inflation-Protected Securities Thursday, February 23, 2023

Treasury Inflation-Protected Securities, or TIPS, are securities whose principal is tied to the Consumer Price Index (CPI). The principal increases with inflation and decreases with deflation. When the security matures, the U.S. Treasury pays the original or adjusted principal, whichever is greater. TIPS pay interest every six months. The accrued principal assumes an initial investment of $1,000. Figures after periods in bid and ask quotes represent 32nds; 101.26 means 101 26/32, or 101.8125% of 100% face value; 99.01 means 99 1/32, or 99.03125% of face value.

MATURITY	COUPON	BID	ASKED	CHG	YIELD*	ACCRUED PRINCIPAL
2023 Apr 15	0.625	100.11	100.13	unch.	-2.308	1195
2023 Jul 15	0.375	100.10	100.12	+2	-0.585	1276
2024 Jan 15	0.625	98.26	98.28	unch.	1.897	1272
2024 Apr 15	0.500	98.02	98.04	unch.	2.104	1177
2024 Jul 15	0.125	97.23	97.25	+1	1.748	1250
2024 Oct 15	0.125	97.05	97.07	unch.	1.857	1157
2025 Jan 15	0.250	96.22	96.24	unch.	2.007	1253
2042 Feb 15	0.750	85.22	85.27	+10	1.620	1314
2043 Feb 15	0.625	82.25	82.30	+10	1.629	1291
2044 Feb 15	1.375	95.00	95.05	+11	1.649	1274
2045 Feb 15	0.750	83.09	83.14	+12	1.651	1261
2046 Feb 15	1.000	87.13	87.18	+11	1.653	1253
2047 Feb 15	0.875	84.10	84.16	+11	1.662	1230
2048 Feb 15	1.000	86.17	86.22	+13	1.653	1204
2049 Feb 15	1.000	86.11	86.16	+13	1.640	1180
2050 Feb 15	0.250	70.11	70.16	+12	1.601	1155
2051 Feb 15	0.125	67.04	67.09	+13	1.576	1140
2052 Feb 15	0.125	66.28	67.00	+12	1.539	1066

*-Yld. to maturity on accrued principal.

Source: Tullett Prebon

Details aside, the TIPS product is best for those investors looking to get protection from long-term CPI-U inflation increases. Also, we can use the WSJ quotes in the exhibit to estimate the market's expected inflation rate.

To first approximation, the TIPS yield resembles a real yield. From the treasury bond report the same day, we can take the nominal yield of the 30-year T-bond to be 3.875%. The real yield for a 30-year TIPS product is estimated at 1.539%, from the last row in the table above . Taking the expected inflation rate equal to the difference, we see that the 30-year expected CPI-U inflation rate is being priced at 2.236%.

It seems investors are optimistic about the U.S. Government's ability to control inflation. Like all things in finance, only the future will tell.

4.6 FLOATING RATE NOTES

Unlike fixed rate bonds, FRNs pay a quarterly variable coupon determined in arrears, implying that the first coupon payment is always known, but later payments are unknown until 3 months before each payment is made. The FRN coupon rate is the sum of the 13-week Treasury-bill auction rate (the index determination rate) and a spread determined by the auction process on the issue date. The spread is constant over the life of the FRN while future variable payments are determined by the fluctuations in the index.

Because the interest payment includes a floating component and a fixed component, it is not a pure floating instrument as described in Chapter 3. The instrument should be valued as a combination of a floating rate note and a fixed rate note. For example, the 2/23/23 FRN auction settled with a high discount margin of 0.200% and an index determination rate of 4.575%. This means that the total annual coupon is the sum, 4.775%. The actual floating index changes weekly. However, after every 3 months until maturity, a payment is made and the following payment is determined using the contemporaneous index determination rate plus the constant spread.

To value a floating rate note accurately, let's consider the auctioned note just mentioned. Suppose we have already completed a ZCB yield curve exercise and have values of r(t) for all values of t. Then, for a \$1000 face value bond, we have a quarterly fixed payment of $0.200\% \times 1000/4 = \$0.50$. The first payment date (t_1) is known, and so is the first payment amount, or $4.575\% \times 1000/4 = \$11.4375$. The present value of the first payment is therefore the sum, \$11.9375, discounted at the t_2 spot rate, i.e. multiplied by $\exp(-r(t_1)t_1)$.

The second payment date (t_2) is also known. We don't know the payment yet, but we can use the forward rate from t_1 to t_2 as a risk-neutral substitute for the expectation. Then the value of the second payment is given by $(0.50 + f(t_1,t_2)*1000/4)$ discounted by multiplying by $\exp(-r(t_2)t_2)$.

Let's perform the valuation using the 2/8/23 ZCB term structure we estimated earlier in the chapter. At the inception of the note, the timing of the payments in years is {0.25,0.50, … 2.00}. The spot discount factors are computed using $d_t = \exp(-r(t)t)$. The forward rates are easily computed from the spot rates:

$$f(t_1, t_2) = \ln\left(\frac{d_1}{d_2}\right)/(t_2 - t_1)$$

The fixed component of cash flow is risk-free, and the risk-neutral floating component can be computed from the forward rates, multiplied by 1000/4. The total risk-neutral cash flow (RNCF) is discounted at the spot rates to find the present value. The result is remarkably close to the face value of \$1000.

Date	2/23/2023					
Spread	0.2000%		**FRN Valuation on issue date**			
Index	4.5750%		**Using 2/8/23 ZCB term structure**			
Notional	1000		*No accrued interest*			

t	Spot d(t)	Forw	Fixed	RN(Float)	RNCF	PV(RNCF)
0.25	0.9867		0.5	11.4375	11.9375	11.7784
0.50	0.9742	0.0509	0.5	12.7183	13.2183	12.8773
0.75	0.9625	0.0484	0.5	12.0880	12.5880	12.1160
1.00	0.9515	0.0461	0.5	11.5210	12.0210	11.4376
1.25	0.9411	0.0440	0.5	11.0117	11.5117	10.8331
1.50	0.9312	0.0422	0.5	10.5553	11.0553	10.2944
1.75	0.9218	0.0406	0.5	10.1475	10.6475	9.8145
2.00	0.9128	0.0391	1000.5	9.7842	1010.2842	922.1825
					PV	1001.3338

In this case, there is no accrued interest, since the pricing is at issuance. However, it is a simple matter to get to the quoted FRN value by deducting the accrued interest.

Eurodollar bonds and LIBOR

FRNs also trade outside the U.S. in the Eurodollar market, using a structure similar to the U.S. Treasury FRNs.[11] In the past, issuers used LIBOR, or the London Interbank Offer Rate for US dollars as a reliable index and benchmark of the cost of short-term dollars. LIBOR also had the benefit of longer maturities being available, particularly 1-month, 3-month, 6-month and 12-month LIBOR. In fact, the 3-month LIBOR was so popular it was chosen as the benchmark for the Eurodollar futures contract by the Chicago Mercantile Exchange. Note that since LIBOR represented short-term unsecured dollar debt between London banks, part of the rate could be attributed to the credit risk of the borrowing bank. Hence, it was not a truly riskless benchmark dollar rate, since it includes a credit risk premium.

Regrettably, in 2012, "multiple criminal settlements by Barclays Bank revealed significant fraud and collusion by member banks connected to the [LIBOR] rate submissions, leading to the scandal."[12] . The LIBOR index is now managed by ICE, the Intercontinental Exchange, and was scheduled to be discontinued in 2021 or 2022. A new standard, the *Secured Overnight Funding Rate* (SOFR) has been recommended as an alternative benchmark. See https://www.financialresearch.gov/data/libor-alternatives/ for details.

According to Rosenblum and Ventura (Goldberg Kohn, 2023), the current situation for legacy LIBOR-based transactions can be summarized as follows:

- As of 12/31/21, no new LIBOR linked contracts could be signed with expiration dates later than 6/30/23
- SOFR is a widely accepted alternative to LIBOR.
- Since it is secured, credit risk is minimal; it is nearly a risk-free rate.
- The SOFR index is based on transactions exclusively, and does not rely on subjective input.
- It is estimated that after 6/30/23, $74 trillion of LIBOR-based contracts will still be in existence, mostly interest rate derivatives.

[11] The "Euro" component of "Eurodollar" has little to do with Europe. Any dollar-denominated instruments traded outside the U.S. are considered Eurodollar investments.
[12] See the Wikipedia entry "LIBOR scandal" for more details

- The LIBOR Law (3/15/22) regulates how disputes will be settled as LIBOR is legally replaced by SOFR plus a spread for the purpose of computing contract payments and receipts.

4.7 MUNICIPAL BONDS

Municipal bonds (munis) are issued by non-federal U.S. government entities, such as states, cities, and their agencies. For example, the Port Authority in New York has issued *authority bonds* to fund tunnels and bridges. These bonds are not guaranteed by the full faith and credit of the U.S. government, so default is a possibility. States and cities have tax authority, but do not have the power to create U.S. dollars. However, in many cases, municipal bonds are rated AAA due to the quality of the underlying collateral, the assumed ability of the municipality to collect taxes and usage fees, or insurance provided by third-party insurance companies.

We will study AAA munis so that we can focus more on implied tax rates as opposed to default risk. The following summary of Muni AAA rates on 2/24/23 was supplied by FMS Bonds Inc. Because municipal bonds trade unevenly in over-the-counter markets, reliable quotes are difficult to find. Nevertheless, this will give us some information.

AAA RATED MUNI BONDS

ISSUE	MATURITY RANGE	TODAY	LAST WEEK
National	10 Year	2.50	2.25
National	20 Year	3.30	3.10
National	30 Year	3.50	3.30

Source: FMS Bonds Inc. 2/24/23

The yields on AAA municipal bonds are lower than the yields on U.S. government bonds. For lower-rated bonds, we would expect that the yields will be higher to reflect some element of default risk and losses given default. However, favorable tax treatment allows muni yields to be lower. From Investopedia.com:

> "While interest income is usually tax-exempt for municipal bonds, capital gains are usually not tax-exempt. Capital gains realized from selling a bond are subject to federal and state taxes. ... These investors are subject to taxation on the interest income from some munis."

Assuming bonds are held to maturity, then, municipal coupons will be untaxed whereas U.S. government bond coupons are taxed at the federal level. The spread between the bonds would then be derived from the GVE:

$$\text{Municipal yield} =$$

$$= (1-\tau_e)\,(\text{Treasury yield} + \text{Expected loss rate} + \text{Additional risk premium})$$

Note that municipal bonds, other things equal, will attract investors in the highest tax brackets. At the time of the FMS chart above, U.S. government yields were 3.99%, 4.15% and 3.95% on 10, 20 and 30-year bonds. Assuming AAA bonds have negligible risk, the effective marginal tax rate could be computed for these terms using the equation. For the 10-year bond, $\tau_e = 37.34\%$, which in 2022 would be the 20% federal marginal tax

rate with 17.34% attributable state and local taxes. This might suggest that the marginal investor is in a highly-taxed U.S. state. For the 20-year bond, τ_e = 20.48%, perhaps suggesting that the marginal investor is in a tax-free state such as Florida or Texas! For the 30-year bond, the implied state/local tax rate is negative.

Clearly, our simple model is not capturing the complexity of this valuation problem, since we are not able to easily assess risks such as changes in the tax code and tax rates over the longer term. The generally declining values of τ_e do seem to be consistent with the idea that the expected tax benefits of municipal bonds decline with time to maturity.

4.8 CORPORATE BONDS

U.S. corporate bonds outstanding totaled about $10 trillion at the end of 2021, according to SIFMA, amounting to roughly 45% of the size of the marketable U.S. government bond market. The structure of corporate fixed rate bonds resembles that of the U.S. treasury bonds, except that some corporations make coupons payments quarterly and some make coupon payments semiannually. However, unlike U.S. government debt, we realize that corporations can default on principal and interest, and creditors are not guaranteed to recover their losses.

Corporate bond pricing models range from simple to complex. A simple reduced-form framework for valuing corporate bonds was introduced by the Bank for International Settlements (BIS) which provides regulatory guidance to financial institutions and governments globally. These standardized abbreviations are well-known by bankers, and we will use the Greek letters for formulas.

- PD = π = Probability of default per year
- LGD = λ = Loss given default, the percentage of a bond's market value that is lost, as of the time of default

To make the framework more realistic, we will allow a risk premium (θ) for a corporate bond in addition to compensation for expected losses. We will also continue to use "r" as the riskless interest rate, and "k" to represent the number of coupon payments per year. If the conditional probability of default is constant from one period to the next, and the LGD is assumed constant, we can easily price corporate bonds using the tools of Chapter 2.

To make this as simple as possible, let's adapt the Excel PV function for an annuity with a final payment, =PV(rate,nper,pmt,fv). Now divide the cash flows from the corporate bond into two parts. The first part consists of payments all equal to ($1 - \lambda$) times the promised values. Since λ is the loss rate, $1 - \lambda$ is the *recovery rate*. Since this amount is assumed to be recovered, it may be considered risk-free. The remaining fraction of the promised payments λ is subject to a constant default rate of π per year. Since nothing is recovered from these cash flows in default, we can value the promised payments by discounting at $r^* = (r + \theta + \pi)/(1 - \pi)$. The resulting valuation can be stated in Excel terms as:

$$-V_0 = (1 - \lambda)PV\left(\frac{r + \theta}{k}, Nk, \frac{cF}{k}, F\right) + \lambda\, PV\left(\frac{r^*}{k}, Nk, \frac{cF}{k}, F\right)$$

Example

VFE Corp's 6% coupon 5-year bond is estimated to have an annual default rate of 3.77%. The LGD is unknown, and therefore assumed to be the standard average value of 45%. The risk-free rate of interest is 5%, and the additional risk premium historically required for VFE Corp's debt has been 3%. The bond makes quarterly coupon payments. What is the value of the bond and its YTM?

Solution

Using the formula above, we have $V_0 = 0.55(918.24) + 0.45(780.63) = 505.03 + 351.28 = \$856.32$. The YTM is found with the IRR function and the promised coupon & principal payments. In this case, YTM = 9.66%.

This can be compared to the brute force calculation done in Excel:

VFE Corp 6% coupon 5-year note

Term	5	years
Payments per year	4	
Face value	1000	
Coupon rate	6.00%	
Actual coupon	15	
Risk-free rate	5.00%	annual
	1.25%	periodic
Pi	3.77%	annual
	0.94%	periodic
Lambda	45.00%	
Theta	3.00%	annual
	0.75%	periodic
r*	2.97%	

Quick Approximation of YTM	
Risk-free (r)	5.00%
Expected loss rate ($\pi\lambda$)	1.70%
Risk premium (θ)	3.00%
Sum	9.70%

Payment	Promised	Recovered	At risk
1	15	8.25	6.75
2	15	8.25	6.75
3	15	8.25	6.75
4	15	8.25	6.75
5	15	8.25	6.75
6	15	8.25	6.75
7	15	8.25	6.75
8	15	8.25	6.75
9	15	8.25	6.75
10	15	8.25	6.75
11	15	8.25	6.75
12	15	8.25	6.75
13	15	8.25	6.75
14	15	8.25	6.75
15	15	8.25	6.75
16	15	8.25	6.75
17	15	8.25	6.75
18	15	8.25	6.75
19	15	8.25	6.75
20	1015	558.25	456.75

Discount rate (qtrly)	2.00%	2.97%
PV	\$505.03	\$351.28
Value		\$856.32
YTM		9.66%

In practice of course, one would have to be concerned with the accrued interest adjustment and the variable risk-free discount rates that were explained in the section on U.S. government bonds.

The table also shows a famous YTM approximation that is popular among bond traders and analysts to decompose the YTM of a corporate bond. There are several ways to reach this approximation. We could compute the LGD-weighted YTM of the two components of the bond valuation formula for example:

$$Approx\ YTM = (1-\lambda)(r+\theta) + \lambda\left(\frac{r+\theta+\pi}{1-\pi}\right) \approx r + \theta + \pi\lambda$$

This approximation works best when π is small. The term $\pi\lambda$ is often called the *expected loss rate*. Therefore, the total YTM of a corporate bond can be represented (approximately) as the sum of the risk-free interest rate for a government bond of similar maturity, the expected loss rate of the bond, and a bond-specific additional risk premium.

Bond traders also split the total YTM into the risk-free rate and the *credit spread*. Using this nomenclature, the credit spread is also the expected loss rate plus the bond-specific risk premium. For investment grade bonds, the expected loss rate is quite small, so the credit spread is dominated by the market risk premium term.

In practice, it is of course useful to be able to value bonds for different sets of parameters. However, we can also move backward. Suppose would like to learn about a company's implied credit quality from the market value of the bonds it has issued. In this case, we know the YTM and simply need to back out the components.

Here are quotes for the most actively traded corporate bonds as of 3/1/23 from FINRA:

Most Active Investment Grade Bonds

Issuer Name	Symbol	Coupon	Maturity	Moody's®/S&P	High	Low	Last	Change	Yield%
AMGEN INC	AMGN5542113	5.250%	03/02/2030	Baa1/BBB+	100.52300	98.82500	99.12300	-0.299000	
AMGEN INC	AMGN5542390	5.250%	03/02/2033	Baa1/BBB+	99.94768	98.45600	98.70500	-0.460000	
CIGNA GROUP	CI4866403	3.750%	07/15/2023	Baa1/A-	99.79433	99.27100	99.43200	0.110800	5.302697
AETNA INC NEW	CVS4370492	2.800%	06/15/2023	Baa2/BBB	99.40300	99.15300	99.24600	-0.093000	5.469234
PEPSICO INC	PEP4551335	3.000%	10/15/2027	A1/A+	94.05900	92.57800	92.67200	-0.387000	4.786204
AT&T INC	T4763018	4.100%	02/15/2028	Baa2/BBB	96.26000	94.31430	94.64600	-0.719700	5.344739
U S BANCORP MEDIUM TERM NTS-BOOK ENTRY	USB4792226	3.375%	02/05/2024	A2/A+	99.21100	97.99100	98.18300	0.028000	5.418111
AMGEN INC	AMGN5542392	5.650%	03/02/2053	Baa1/BBB+	98.95900	97.83100	98.40500	-0.682000	
META PLATFORMS INC	FB5522214	4.450%	08/15/2052	A1/AA-	83.19800	81.87300	82.10700	-0.544000	5.712253
COMCAST CORP NEW	CMCS5406980	2.887%	11/01/2051	/A-	64.47200	63.82800	63.82800	-0.673000	5.374975

Investment grade bonds are those higher in credit quality than other bonds, with a minimum rating of Baa/BBB on Moody's/S&P bond rating services. Bonds with lower ratings are called *high yield* bonds by marketers and *junk bonds* by investors.

Ratings are chosen by the ratings agencies to give their best possible estimate of the likelihood of default of the bond. They do not attempt to measure loss given default, but estimates of LGD can be built from models or purchased from analytic firms. This S&P default study was published in April 2022:

Table 26

Global Corporate Average Cumulative Default Rates By Rating Modifier (1981-2021)

(%)	--Time horizon (years)--														
Rating	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
AAA	0.00	0.03	0.13	0.24	0.34	0.45	0.50	0.58	0.64	0.69	0.72	0.75	0.78	0.83	0.89
AA+	0.00	0.05	0.05	0.09	0.14	0.19	0.25	0.30	0.35	0.41	0.46	0.52	0.58	0.64	0.70
AA	0.02	0.03	0.08	0.21	0.34	0.46	0.58	0.69	0.77	0.86	0.93	0.99	1.08	1.14	1.20
AA-	0.03	0.08	0.16	0.23	0.30	0.40	0.46	0.51	0.55	0.60	0.66	0.71	0.73	0.77	0.81
A+	0.05	0.08	0.18	0.30	0.39	0.48	0.58	0.68	0.80	0.93	1.04	1.17	1.31	1.47	1.62
A	0.05	0.13	0.21	0.31	0.43	0.59	0.75	0.90	1.07	1.27	1.42	1.54	1.65	1.72	1.87
A-	0.05	0.15	0.24	0.34	0.48	0.63	0.83	0.98	1.10	1.20	1.30	1.42	1.54	1.65	1.74
BBB+	0.09	0.25	0.45	0.64	0.85	1.09	1.28	1.49	1.75	1.99	2.21	2.36	2.53	2.75	2.98
BBB	0.14	0.36	0.56	0.89	1.21	1.54	1.86	2.15	2.45	2.75	3.07	3.32	3.54	3.64	3.83
BBB-	0.23	0.66	1.23	1.87	2.53	3.13	3.63	4.10	4.48	4.83	5.20	5.52	5.81	6.26	6.60
BB+	0.31	0.94	1.69	2.44	3.19	3.95	4.59	5.06	5.62	6.18	6.57	7.03	7.50	7.85	8.39
BB	0.46	1.45	2.82	4.12	5.47	6.57	7.58	8.46	9.32	10.06	10.85	11.43	11.90	12.22	12.56
BB-	0.92	2.86	4.89	6.97	8.79	10.54	12.02	13.41	14.53	15.48	16.22	16.96	17.67	18.34	18.97
B+	1.94	5.31	8.64	11.48	13.73	15.46	17.02	18.37	19.60	20.73	21.70	22.39	23.13	23.80	24.45
B	2.99	6.99	10.59	13.54	15.91	18.05	19.56	20.69	21.80	22.91	23.64	24.37	24.92	25.37	25.87
B-	5.89	12.59	17.88	21.56	24.26	26.23	27.69	28.84	29.65	30.34	31.41	32.10	32.71	33.34	33.77
CCC/C	26.55	36.74	41.80	44.74	46.91	47.95	49.08	49.82	50.48	51.05	51.49	51.92	52.45	52.91	52.97
Investment-grade	0.08	0.23	0.40	0.61	0.83	1.05	1.26	1.45	1.63	1.81	1.98	2.13	2.27	2.40	2.55
Speculative-grade	3.60	6.97	9.86	12.23	14.16	15.75	17.06	18.16	19.14	20.04	20.80	21.44	22.05	22.58	23.09
All rated	1.50	2.93	4.17	5.22	6.10	6.83	7.45	7.97	8.43	8.86	9.23	9.54	9.84	10.10	10.36

Sources: S&P Global Ratings Research and S&P Global Market Intelligence's CreditPro®.

This chart shows the cumulative default rates of global issuers measured from the first ratings date. For example, in the BBB- category (lowest investment grade), the historical one-year default rate was 0.23% or 23 bppa. For a 15-year cumulative period, the default rate was 6.60%, or 44 bppa. For investment grade bonds overall, the one-year default rate was 8 bppa, or (255/15 =) 17 bppa over a 15-year period.

Clearly, there are many assumptions one could make about default probabilities, losses given default and the risk premium, and it is impossible to get estimates of all three from a single spread observation. Nevertheless, skilled corporate bond analysts use all the means at their disposal to estimate these parameters as carefully as possible.

The bottom line: Corporate bond prices hold information about default risk and risk premia that can be useful not only for bond buyers, but for any analyst needing to understand more about the companies they are studying.

Example

Suppose we have a corporate bond that makes 10 annual payments of $50, returning $1000 in principal at the end of 10 years. The risk-free rate is 5%, but the price of this bond is $828.40, not $1000. What can we say about the implied default probability, assuming that the loss given default λ=60%, and there is no additional risk premium?

Solution

The rate that makes the present value of these payments equal to $828.40 is 7.5%. Therefore the "credit spread" is 2.5%. Using the rule of thumb, and assuming no additional risk premium, this is the same as the expected loss premium ($\pi\lambda$), and λ=60%. Therefore π=4.17%, implying an average of about 4.17% expected default rate per year. The exact valuation and yield can be computed using the exact formulae provided above.

4.9 REFERENCES AND ADDITIONAL READINGS

Brumma, Nina, Nunzia Rainone and Richard Crecel, "Global Credit Data LGD Report 2020", May 2020

Garbade, Kenneth, "Why the U.S. Treasury Began Auctioning Treasury Bills in 1929", FRBNY Economic Policy Review, July 2008.

Kraemer, Nick and Jon Palmer, 2021 Annual Global Corporate Default and Rating Transition Study, S&P Global Ratings, April 13, 2022

Rosenblum, James and Kevan Ventura, "The Demise of and Transition from LIBOR", Goldberg Kohn Ltd., January 21, 2023.

Tuckman, Bruce and Angel Serrat, **Fixed Income Securities: Tools for Today's Markets**, Wiley Finance 2022.

4.10 EXERCISES

1. A 5 year idealized U.S. government bond with a 1.75% coupon is valued at 99.81.
    a. What is its YTM (i.e. twice the semiannual yield)?
    b. If the YTM increases by 1%, what is the new price?
2. A Treasury bill with 135 days to maturity has an asking yield of 0.95%.
    a. What is the price of the T-Bill?
    b. What is the effective annual return?
3. Suppose the workers at Gideon Industries negotiated a contact that paid out $10,000,000 per year for the next five years.
    a. If the current risk-free rate is 5%, what would be the present value of a COLA (cost of living adjustment) clause in their contract for the five year period? Use the TIPS data provided in the chapter.

b. What is the fixed wage increase that would make the union indifferent between the COLA and a fixed wage structure?

2. A five-year AAA rated municipal bond yields 1.5% when a five-year government bond yield 2.0%. You have two investor clients, one in the 20% tax bracket and one in a 45% tax bracket. What would you advise them about buying this particular municipal bond compared to the U.S. government bond?

3. Download current TIPS yields and yields for U.S. government bonds. What can you infer about the
   a. Expected inflation rate
   b. Real rate of interest

4. A corporate junk bond is rated CCC by S&P and yields 14% when the yield of a comparable government bond is 2%.
   a. What can you infer about loss given default?
   b. If the additional risk premium is 5%, what can you infer about LGD?

MINIPROJECT 4: THE INFLATION STORY

Following Miniproject 3, we would like to build a ZCB term structure for inflation-indexed bonds. Download TIPS prices and Treasury bond/notes quotes to build a simultaneous nominal ZCB term structure, and real ZCB term structure.

This project should be completed in Python, and you should use APIs to download pricing data directly from the Wall Street Journal or other source. Once completed, you may use your model to build nominal and real term structures for any date, at any time in the future.

The outputs are the same as Miniproject 3. Your deliverable is a Jupyter Notebook file.